

ME5412 Robotics for Healthcare

Lower Limb Rehab Robots

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Content

- Introduction of lower limb rehab robotics
- Design of lower limb rehab robotics
- Control of lower limb rehab robotics
- Advanced topics in lower limb rehab robotics
- Clinical experiments of lower limb rehab robotics
- Challenges and future trends of lower limb rehab robotics

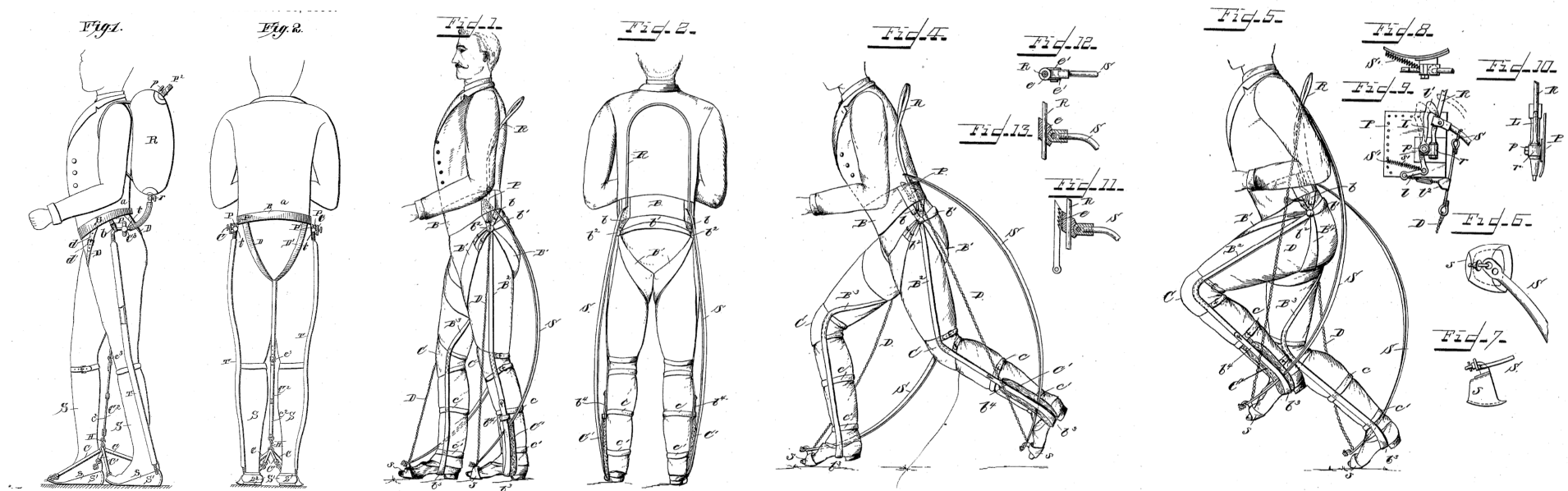
Brief video introduction of lower limb rehabilitation robotics



Video from YouTube channel **Mecharithm-Robotics and Mechatronics**
(<https://www.youtube.com/watch?v=5Qy062PiVHY>)

History of lower limb rehab robotics

● Earliest Devices (1890s)

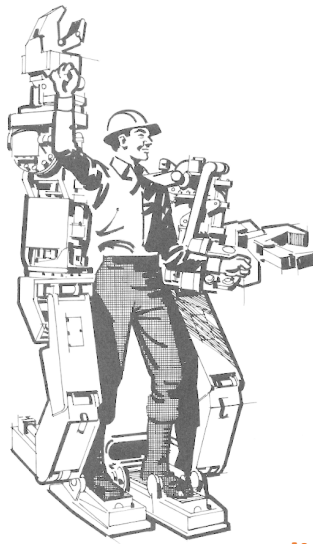


The **earliest** powered and unpowered lower limb rehab designs were patented by Nicholas Yagn in the 1890s.

<https://strydr.co/exoskeletons/>

History of lower limb rehab robotics

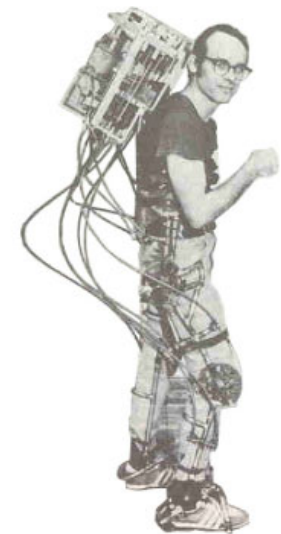
● Early Concepts (1960s-1970s)



Hardiman exoskeleton



Pupin Institute exoskeleton



Wisconsin exoskeleton

The initial ideas for lower limb rehabilitation robotics emerged in the 1960s and 1970s. The US Army/Navy commissioned General Electric Co. in the early 1960s to design and assemble the Hardiman exoskeleton to provide strength augmentation that would allow the operator to lift 680kg.

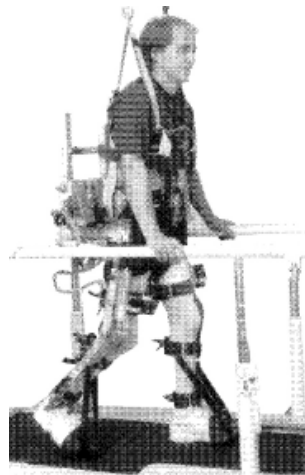
Figure from Wikipedia.

History of lower limb rehab robotics

● Pioneering Devices (1980s-1990s)



The earliest version of Locomat in 1999



The latest version of Locomat in 2021

In the 1980s and 1990s, the field witnessed the development of pioneering devices. Notable among them was the "Lokomat,". The Lokomat was a treadmill-based robotic gait orthosis designed to assist individuals with gait impairments.

History of lower limb rehab robotics

● Advancements in Actuation and Sensing (2000s):



MIT orthosis (2004)
(ankle-foot)



Umich orthosis (2005)
(ankle-foot)



Gravity-balancing orthosis
(lower extremity)



Rehab orthosis (2007)
(knee)



LOPES exoskeleton (2007)
(lower extremity)



ALEX exoskeleton (2009)
(lower extremity)

The early 2000s saw significant advancements in actuation systems and sensing technologies. Robotics researchers began incorporating more sophisticated sensors, such as accelerometers and gyroscopes, to provide better feedback and control. This era marked a shift towards more compact and wearable robotic exoskeletons.

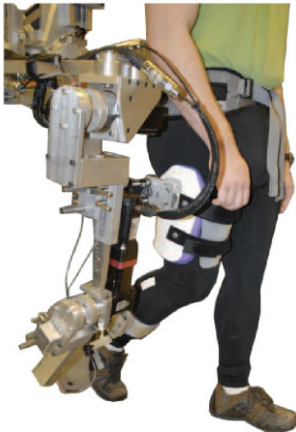
Details of these robotics can be seen in Ding Y. Control and Optimization of soft exosuit to improve the efficiency of human walking[D]. Harvard University, 2018.

History of lower limb rehab robotics

- A surge in development in wearable robotics (2010s)



Vanderbilt orthosis
(lower extremity)



ALEXII exoskeleton
(lower extremity)



Mina orthosis
(lower extremity)



Rewalk orthosis
(lower extremity)



MINDWALKER
(lower extremity)



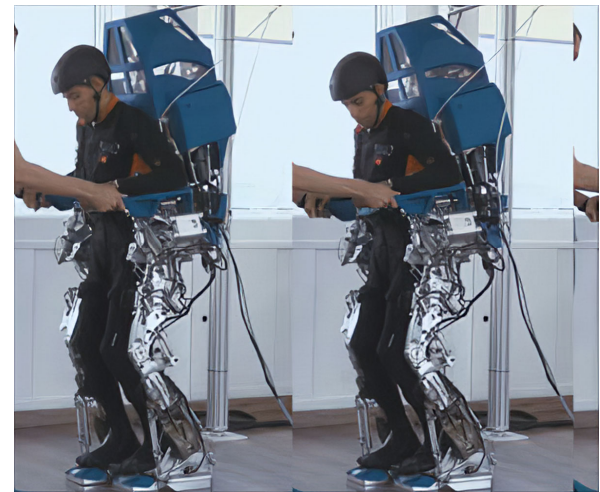
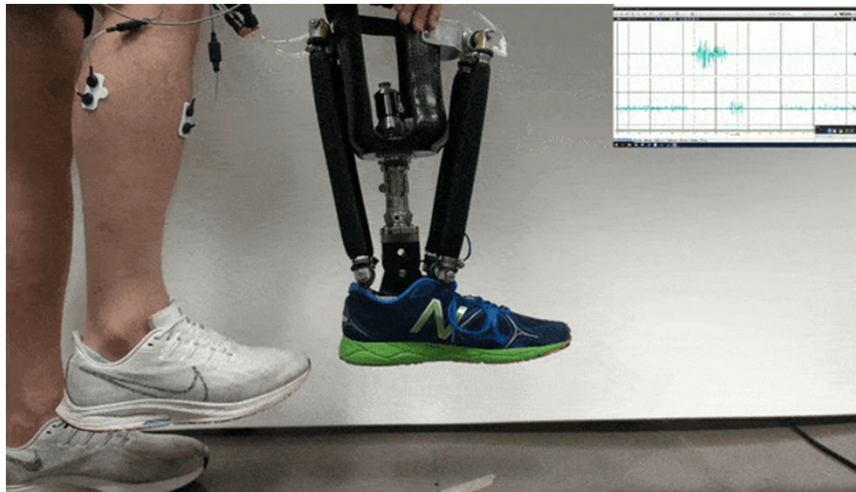
CMU orthosis
(ankle)

The 2010s witnessed a surge in the development of wearable robotic exoskeletons. These devices are designed to be worn by individuals to assist and enhance their lower limb movements. Companies and research institutions worldwide began working on exoskeletons for both clinical rehabilitation and personal use.

Details of these robotics can be seen in Ding Y. Control and Optimization of soft exosuit to improve the efficiency of human walking[D]. Harvard University, 2018.

History of lower limb rehab robotics

- Neurorehabilitation and EMG and Brain-Machine Interfaces (2010s-present)



In recent years, there has been a growing emphasis on incorporating neurorehabilitation principles into robotic devices. Researchers are exploring the integration of brain-machine interfaces (BMIs) to enhance the communication between the human brain and robotic systems, providing more natural and intuitive control.

Categories of lower limb rehab robotics

● Function



Gait rehabilitation



Human locomotion assistance

- **Gait rehabilitation:** the act of learning how to walk after sustaining an injury or disability.
- **Human locomotion assistance:** the use of wearable robotic devices to assist and enhance the ability of individuals to move and walk.

Categories of lower limb rehab robotics

- Passive or Powered



Passive



Powered



- **Passive:** which are not motorized, are often used for ergonomic support, to prevent stress injuries, or to help hold tools or equipment
- **Powered:** use batteries or electric cable connections to run sensors and actuators

Categories of lower limb rehab robotics

- Rigid or Soft



Rigid



Soft

- **Rigid:** both the robot's links and its actuation are much stiffer than the human arm's mechanical properties
- **Soft:** also known as soft wearable robots or exosuits, are made of flexible and compliant materials such as textiles and lightweight fabrics.

Categories of lower limb wearable robotics

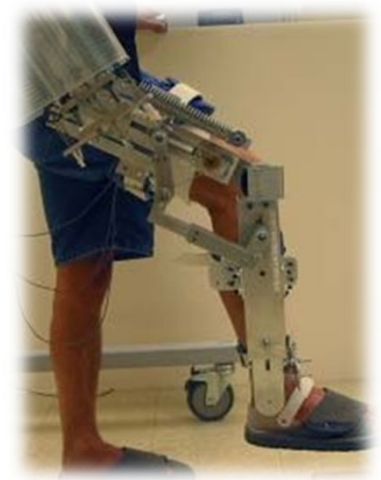
- Full body or partial Exoskeletons



Full body Exoskeletons



Partial Exoskeletons



- **Full body exoskeletons:** is used to describe a device that assistive both upper limb and lower limb of the wearer
- **Partial exoskeletons:** is typically used to describe a device that is used to assist part of the limbs of the wearer

Herr H. Exoskeletons and orthoses: classification, design challenges and future directions[J]. Journal of neuroengineering and rehabilitation, 2009, 6: 1-9.

Categories of lower limb rehab robotics

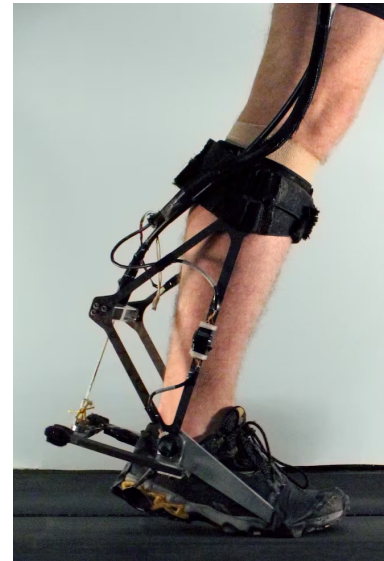
- Hip, knee, ankle, full lower extremity



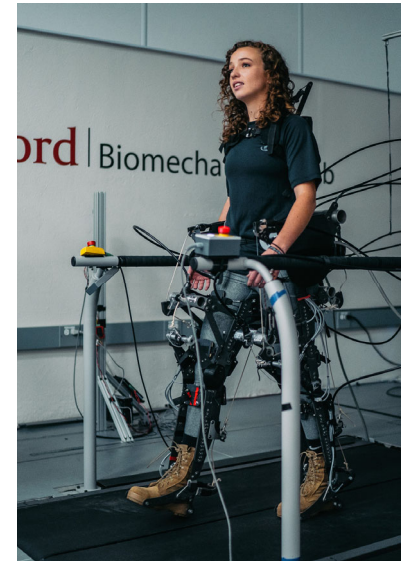
Hip exoskeleton



Knee exoskeleton



Ankle exoskeleton

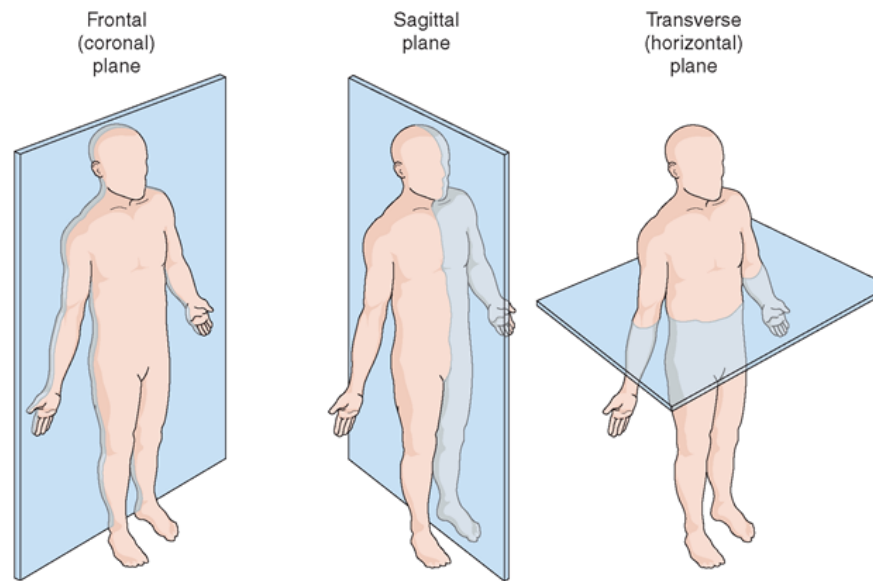


Lower extremity exoskeleton

- This classification is based on the exoskeletons used for which lower limb joints.

Concepts of lower limb movements

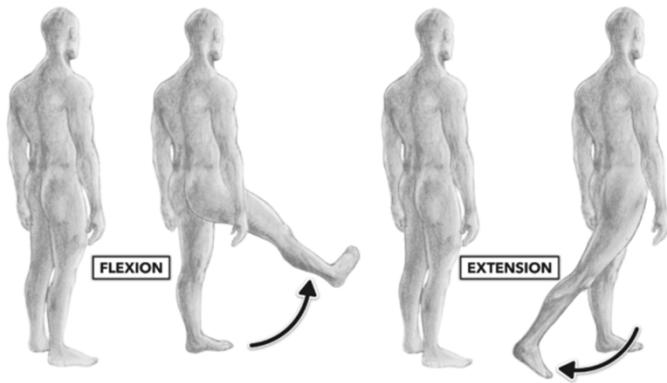
● Human anatomy



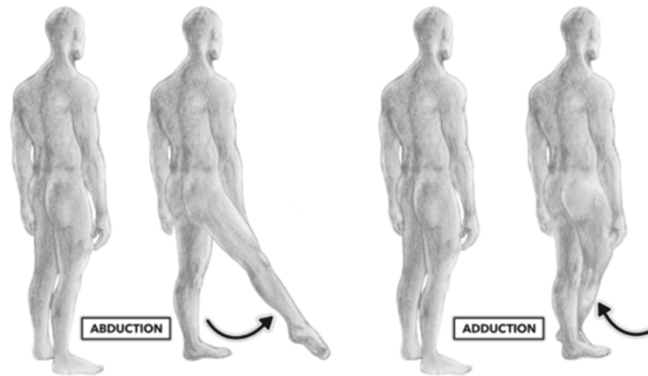
- **Sagittal plane:** Movements that occur within this plane are typically forward-backward movements and include flexion and extension.
- **Frontal plane:** Movements within this plane are usually side-to-side movements and include abduction and adduction.
- **Horizontal plane:** Movements within this plane are typically rotational and include medial and lateral rotation.

Concepts of lower limb movements

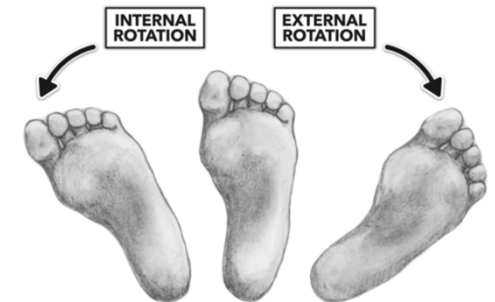
● Hip movements



Flexion and extension



Abduction and adduction

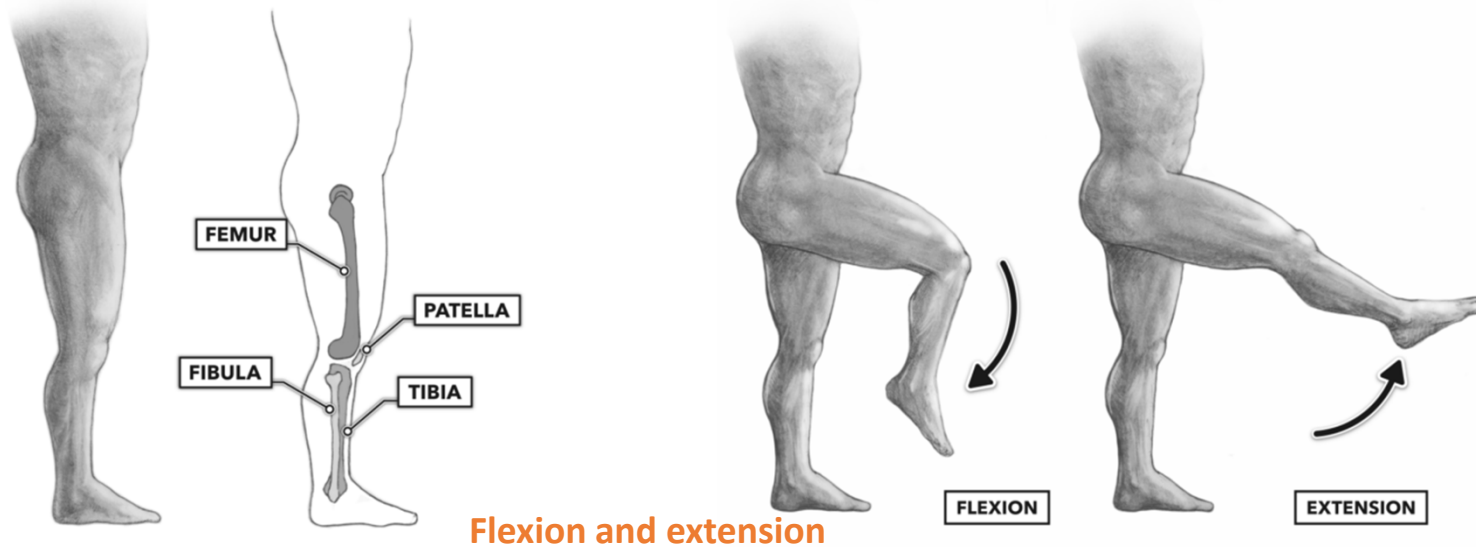


External and internal rotation

- **Flexion and extension:** flexion bring your thigh forward, toward your abdomen; extension is the opposite movement of flexion
- **Abduction and adduction:** move your thigh away from the midline of your body; adduction is the opposite movement of abduction.
- **External and internal rotation:** turn the thigh or leg outward; internal rotation is the opposite movement of external rotation.

Concepts of lower limb movements

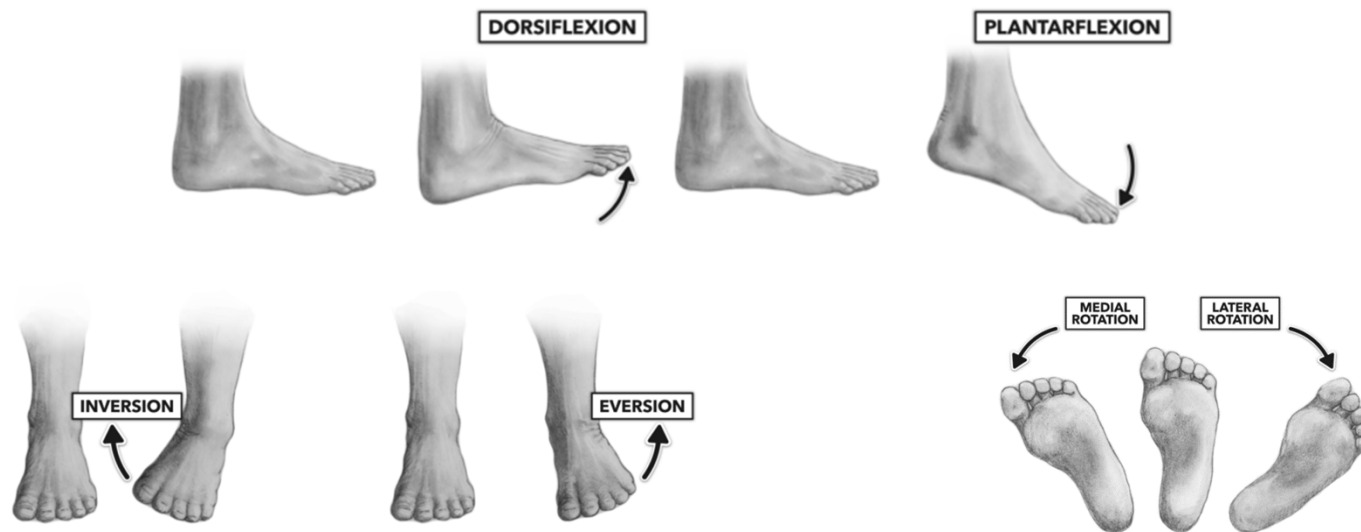
- Knee movements



- **Flexion and extension:** The knee joint is a hinge joint that connects the thigh bone (femur) to the shin bone (tibia) and the smaller bone in the lower leg (fibula). The primary movements at the knee joint are flexion and extension. Knee flexion is the movement that decreases the angle between the thigh and the lower leg, bringing the foot closer to the buttocks.

Concepts of lower limb movements

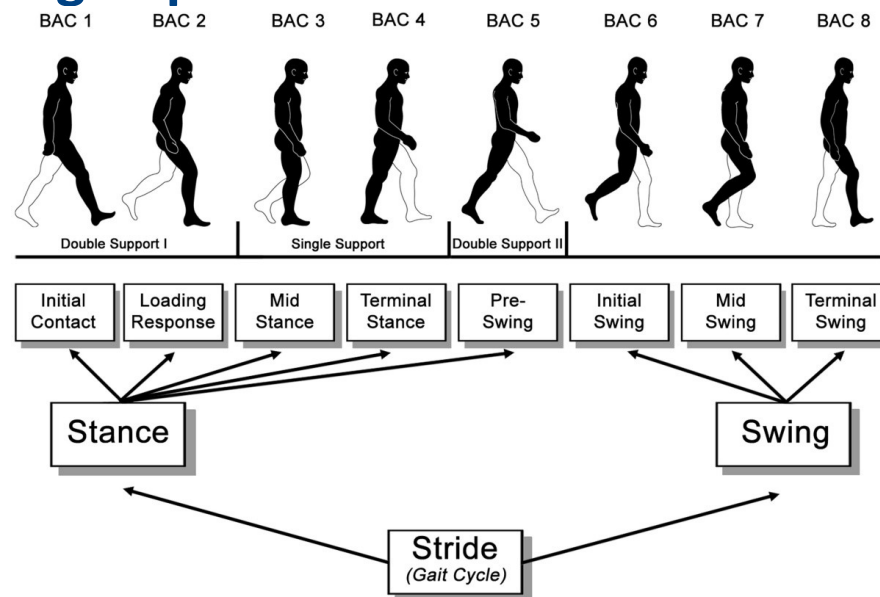
● Ankle movements



- **Dorsiflexion and Plantarflexion:** pulling the toes toward the shin, decreasing the angle between the foot and shin; plantarflexion opposite
- **Inversion and Eversion:** tilting the sole of the foot toward the midline of the body; eversion opposite
- **Medial and Lateral Rotation:** rotating the foot or leg so that the toes point toward the midline of the body; lateral rotation opposite

Concepts of lower limb movements

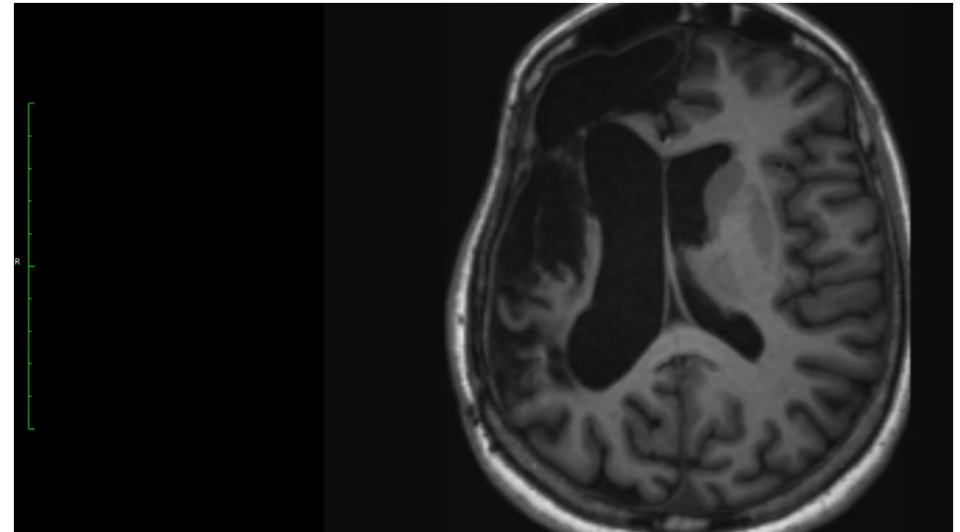
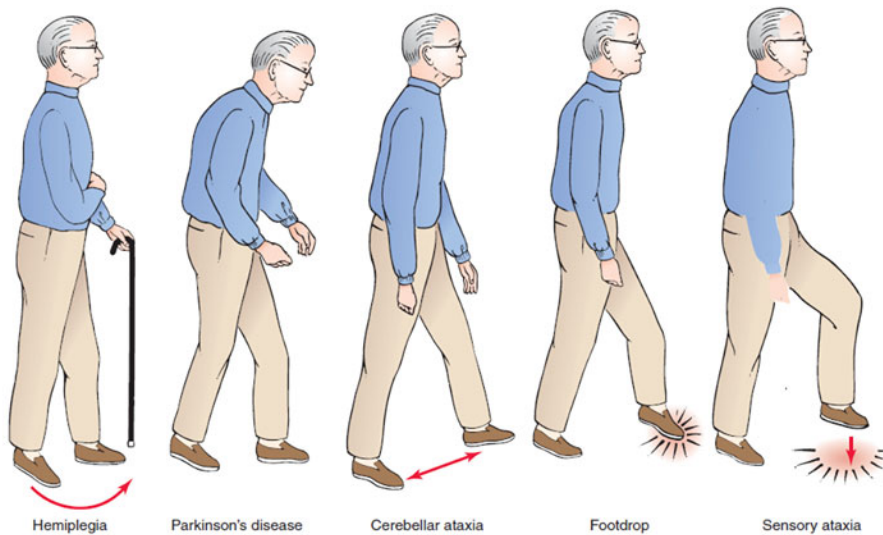
● Human gait and gait phase



- **Gait:** refers to the pattern of movement of the human limbs of animals
- **Gait phase:** a specific period of the walking for an individual. Gait can be broken down into two main phases: stance phase and swing phase. These two phases can be further divided into sub-phases.

Concepts of lower limb movements

● Stroke gait



- **Stroke gait:** also known as hemiplegic gait or post-stroke gait, refers to the way a person walks after experiencing a stroke. A stroke is a medical condition that occurs when there is a sudden interruption in the blood supply to the brain, leading to damage of brain tissue.

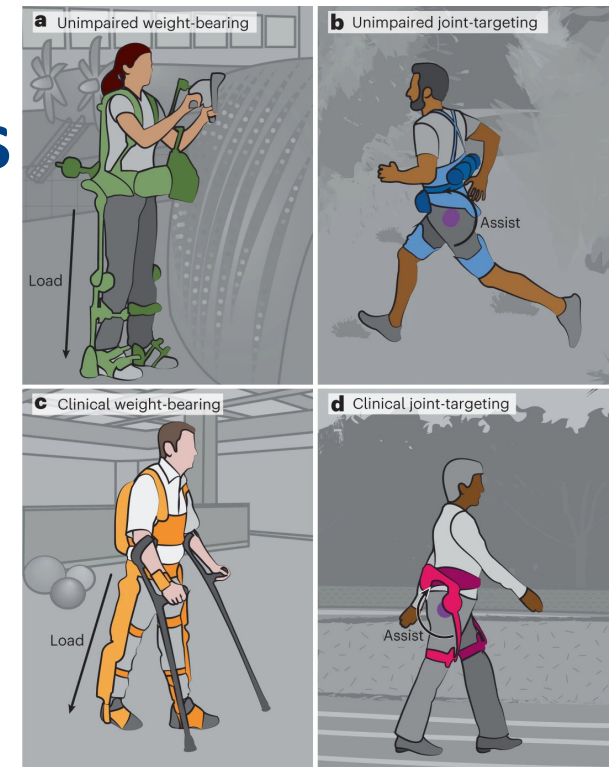
Apart from <https://strokesciences.com/post-stroke-hemiplegic-gait/>

Design of lower limb rehab robotics

● Design principles

The values of joint angle, torque and power in sagittal plane

	Degree (deg)	Torque(Nm/kg)	Power(W/kg)
Hip	-20~+30	-1.0~+1.0	-0.7~+1.5
Knee	0~+50	-0.5~+0.5	-1.5~+1.0
Ankle	-25~+10	0~+1.5	-0.5~+4.0



[Nat. Biomed. Eng., 2023]

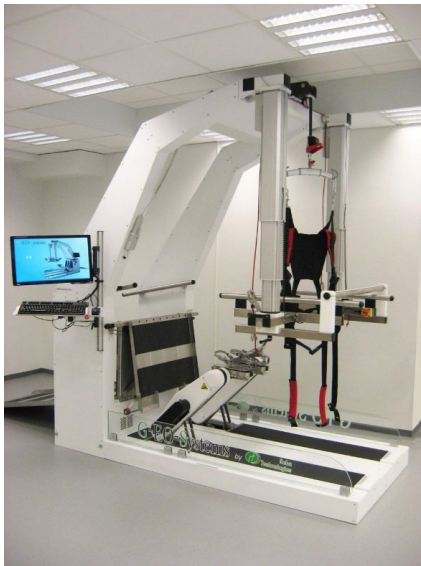
- **Function and use:** fulfill the requirements of rehab, such as: gait training, human locomotion assistance
- **Biomechanical Compatibility:** fulfill the hip, knee and ankle requirements (actuator design: motor selection, transmission design,...)
- **Lightweight and ergonomic design:** be lightweight to minimize user fatigue and facilitate natural movements
- **Adjustability and Customization:** accommodate individual user characteristics such as leg length, joint range of motion, and muscle strength.

Siviy C, Baker L M, Quinlivan B T, et al. Opportunities and challenges in the development of exoskeletons for locomotor assistance[J]. Nature Biomedical Engineering, 2023, 7(4): 456-472.

Design of lower limb rehab robotics

● End effector design

(G-EO system)



- **End effector design:** End effectors are custom engineered for the particular task which is to be performed .
- The G-EO system attaches only to the feet and controls the movements of the foot while the knee, hip and trunk can move more freely during walking.

<https://www.rehatechnology.com/en/products/g-eo-system>

Design of lower limb rehab robotics

- Fixed platform design



(Lokomat)



- **Fixed platform design:** Fixed or stationary exoskeletons systems incorporate a fixed structure combined with a moving ground platform (such as a treadmill)

Design of lower limb rehab robotics

- Mobile exoskeletons design



- **Mobile exoskeletons design:** Most exoskeletons are used for gait rehabilitation or human locomotion assistance. These rehab robotics are mobile devices and can achieve more locomotion tasks compared to fixed platform exoskeletons.

Design of lower limb rehab robotics

● Robotic walker design



Andago (Hocoma, Switzerland)

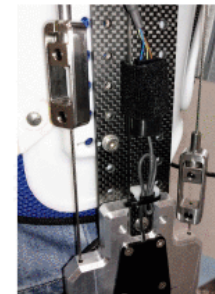
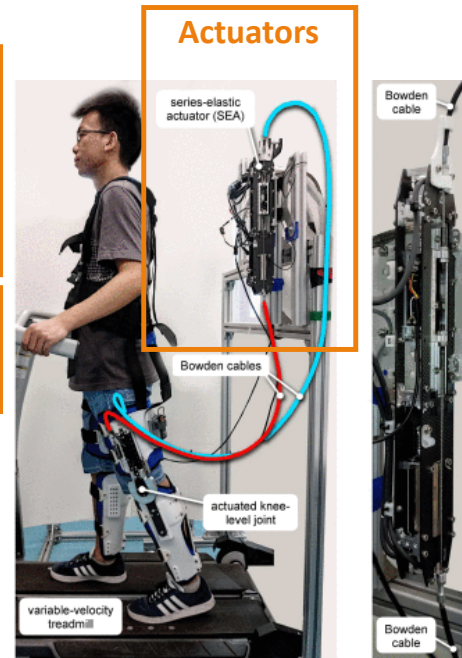
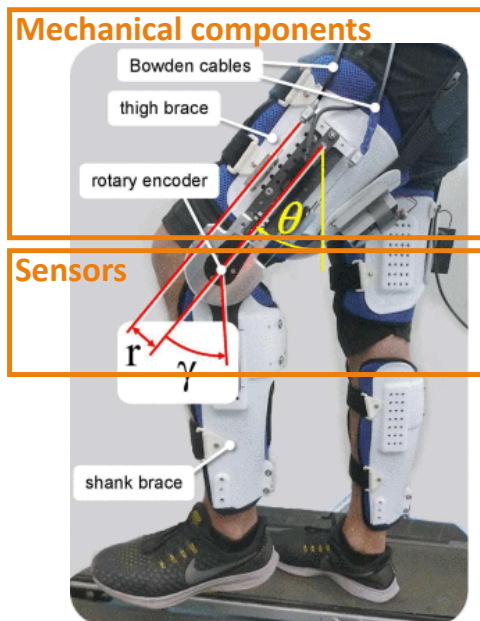


(NUS robotic walker)

- **Robotic walker design:** Robotic walker can be seen as the combination of fixed platform and mobile exoskeleton, which can provide active body weight support, balance training, integrated FES for joint muscle stimulation.

Design of lower limb rehab robotics

- Design components



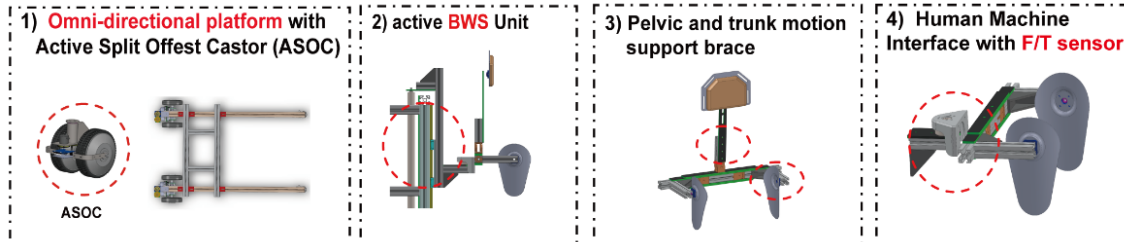
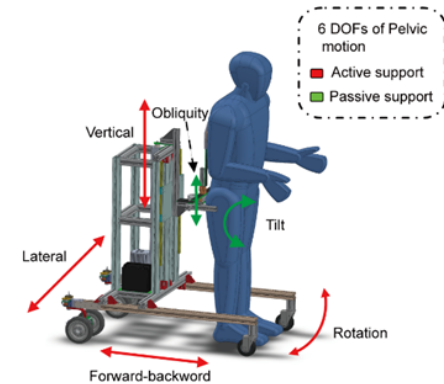
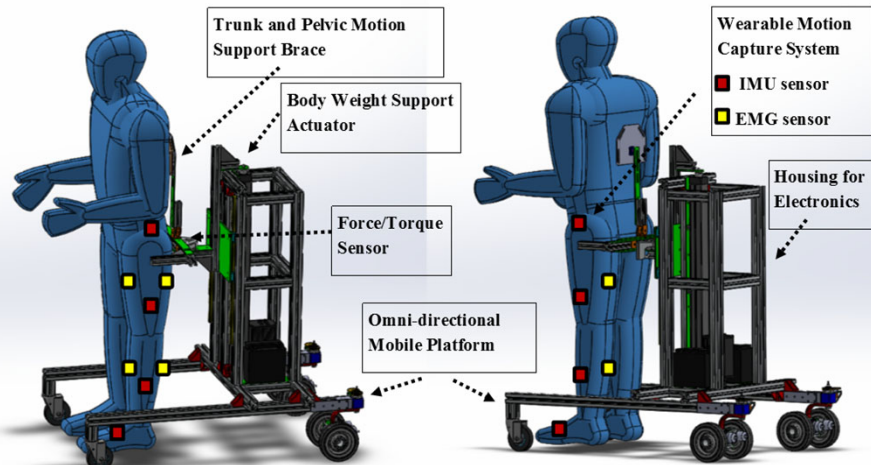
[TRO., 2020]

- Lower limb rehab robotics consist of mechanical links, **joint actuators**, **batteries**, and a series of **sensors**. Actuators are the most important part.

Aguirre-Ollinger G, Yu H. Lower-limb exoskeleton with variable-structure series elastic actuators: Phase-synchronized force control for gait asymmetry correction[J]. IEEE Transactions on Robotics, 2020, 37(3): 763-779.

Design of lower limb rehab robotics

● Design components (robotic walker case study)

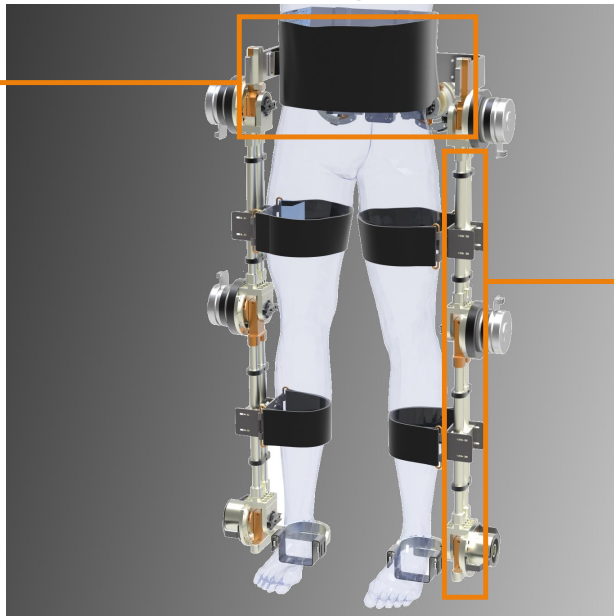


- NUS robotic walker design components includes mechanical structures (such as brace, mobile platform, housing for electronics), actuation system, sensors (such as force/torque sensor, IMU), microcontroller system, and power supply equipment.

Design of lower limb rehab robotics

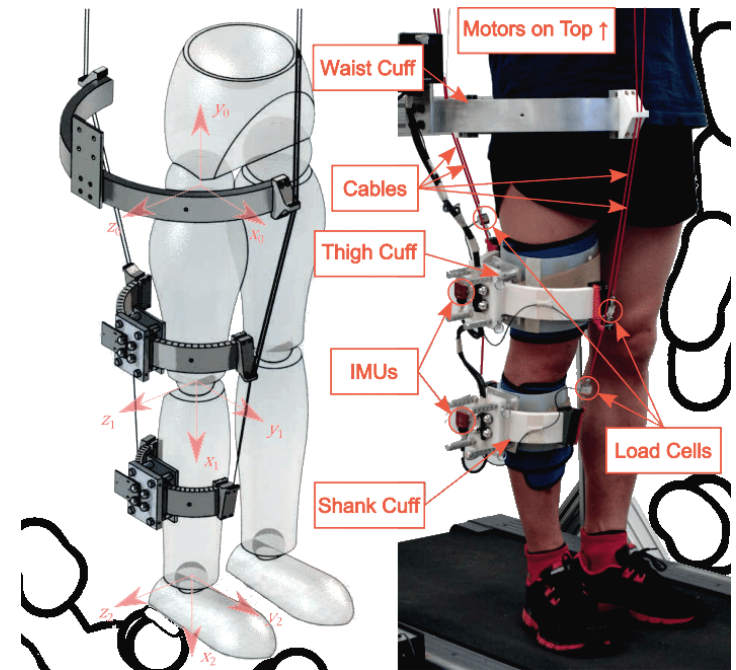
● Mechanical components

Tying portion



Supporting portion

[ETH exoskeletons]



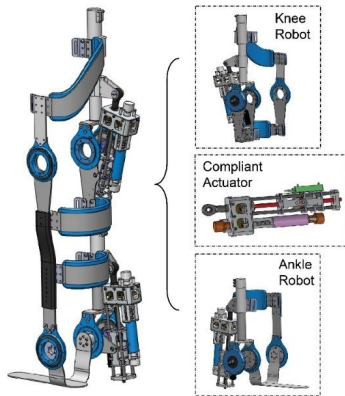
- **Mechanical components:** human body skeleton structures, joint structures, and a flexible driver. The different components of the exoskeleton device include a supporting portion, a cushion, a tying portion, a telescopic rod, and a bearing rod.

Grosu S, De Rijcke L, Grosu V, et al. Driving robotic exoskeletons using cable-based transmissions: a qualitative analysis and overview[J]. Applied Mechanics Reviews, 2018, 70(6): 060801.

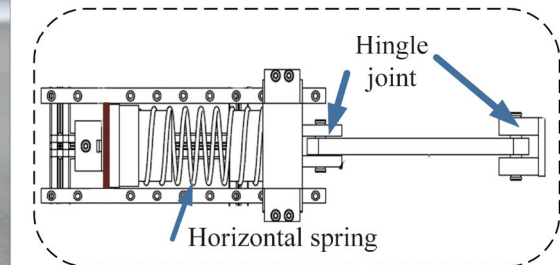
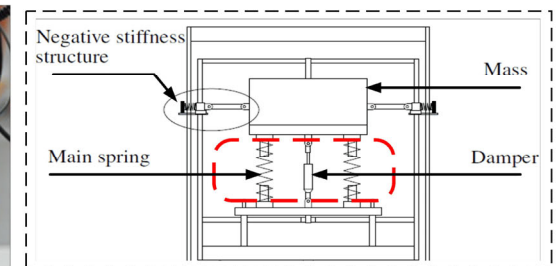
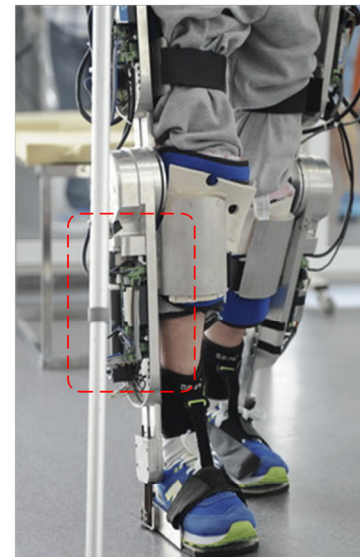
Design of lower limb rehab robotics

● Actuators—Series Elastic Actuator (SEA)

ROBOTICS
*A Portable Knee-Ankle-Foot
Robot for Gait Rehabilitation*



[NUS keen-ankle-foot exoskeleton]

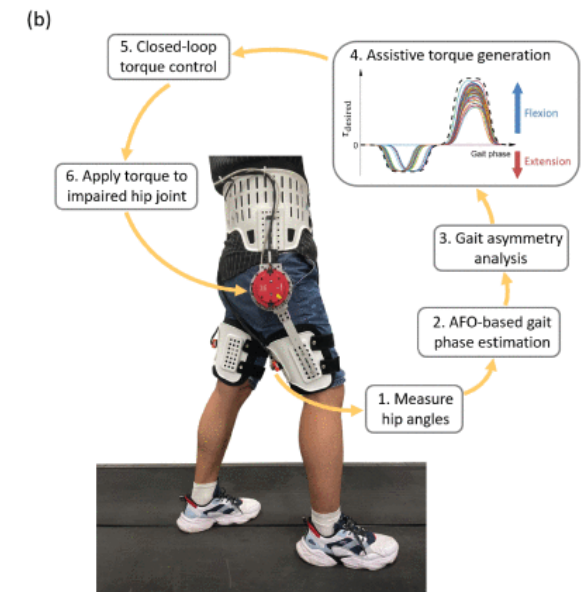
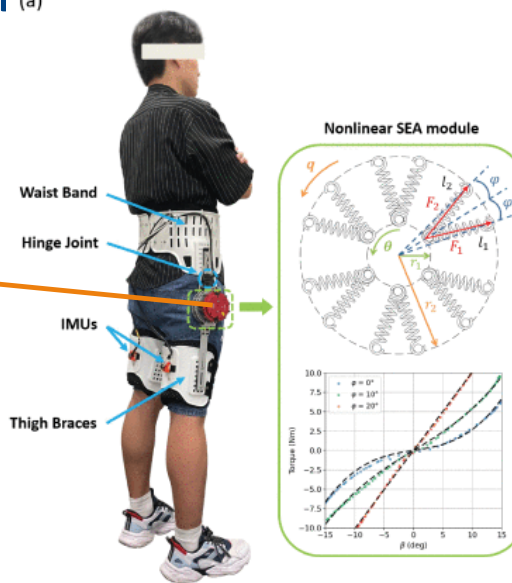
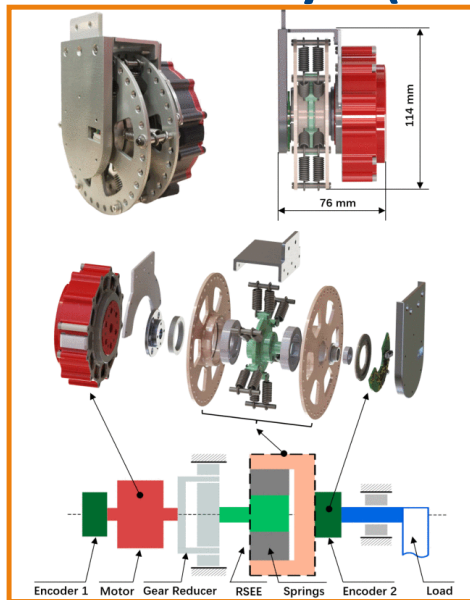


[Actuators, 2023]

- **SEA-driven:** SEA is a type of actuator that incorporates an **elastic element in series** with the motor and gearbox. The primary purpose of the elastic element, which is often a **spring**, is to provide a **compliant interface** with whatever the actuator is driving or manipulating.

Design of lower limb rehab robotics

● Case study—(SEA-driven)^(a)



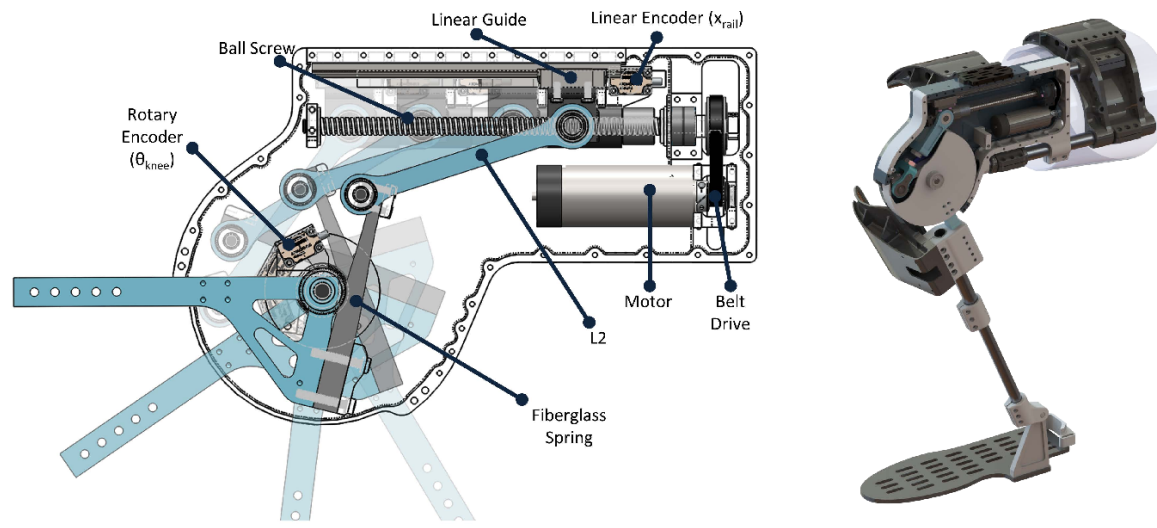
- **Case study:** a novel nonlinear SEA can control the interaction torque with high accuracy and robustness.

The lower interaction torque leading to a negligible negative influence on the hip joint's range of motion, walking speed, and energy expenditure when wearing the hip exoskeleton.

Qian Y, Han S, Wang Y, et al. Toward improving actuation transparency and safety of a hip exoskeleton with a novel nonlinear series elastic actuator[J]. IEEE/ASME Transactions on Mechatronics, 2022, 28(1): 417-428.

Design of lower limb rehab robotics

● Case study—(linear actuator)



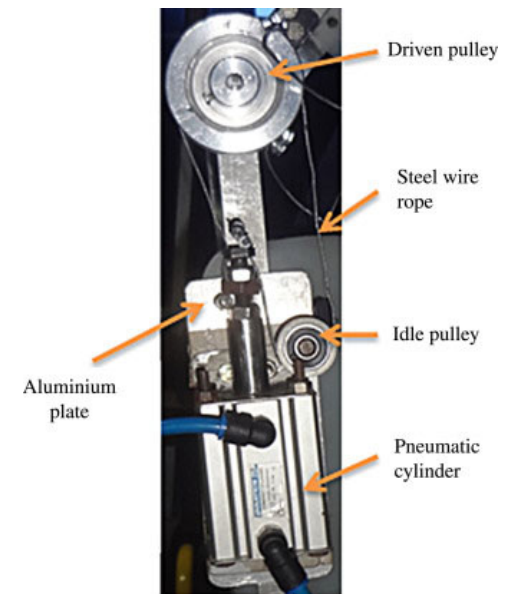
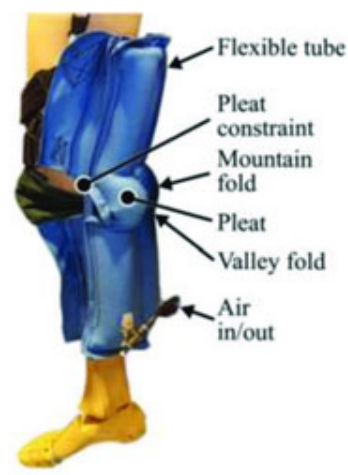
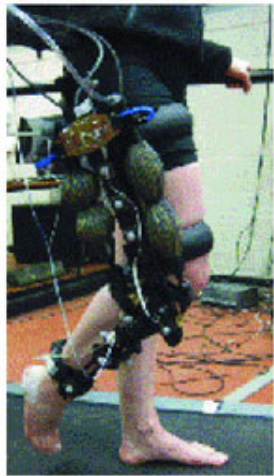
[Northwestern University keen exoskeleton]

- **Case study:** provide knee extension assistance to facilitate sit-to-stand motion of stroke patients; Supply 100% of the required torques and speeds.

Shepherd M K, Rouse E J. Design and characterization of a torque-controllable actuator for knee assistance during sit-to-stand[C]//2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). IEEE, 2016: 2228-2231.

Design of lower limb rehab robotics

● Actuators—Pneumatic Actuators

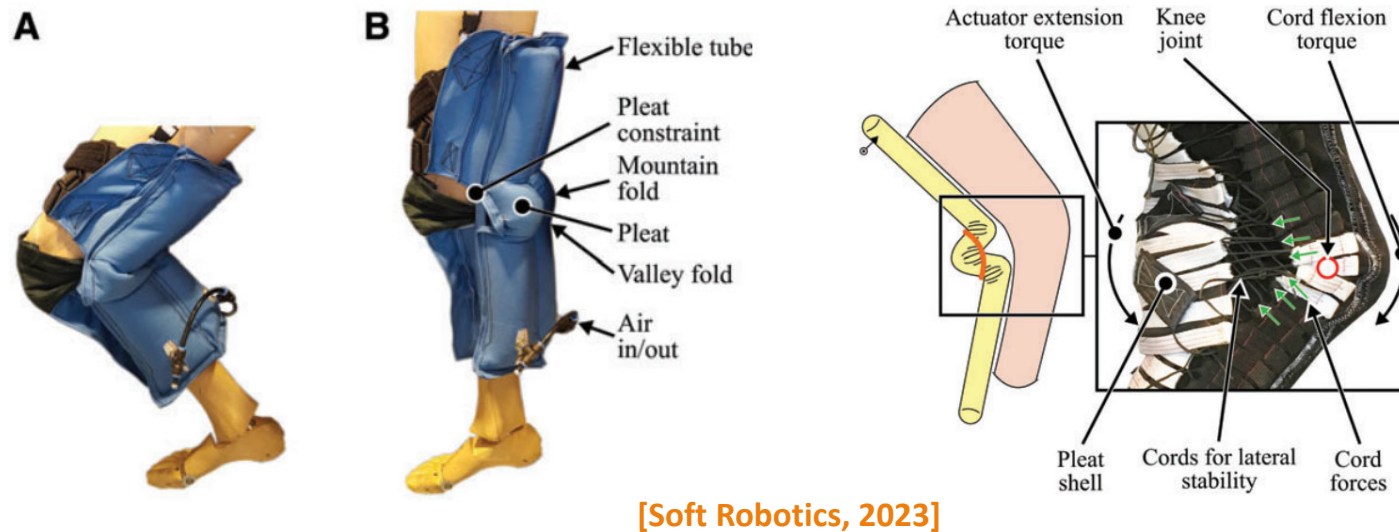


- **Pneumatic actuators:** Pneumatic muscle actuators inflate a sheathed bladder by employing a compressed gas to make it contract lengthwise when expanded radially and have been successfully applied in robotics and robotic therapy.

Figures apart from Wu Z, Yang M, Xia Y, et al. Mechanical structural design and actuation technologies of powered knee exoskeletons: A review[J]. Applied Sciences, 2023, 13(2): 1064.

Design of lower limb rehab robotics

● Case study—(Pneumatic muscle actuators)



- **Case study:** The structure, the pleated pneumatic interference actuator (PPIA), is based on pleated inflatables; is lightweight, collapsible, and clothing integratable; and generates torque from buckling of a constrained fabricreinforced rubber tube. Multiple PPIAs were integrated into a soft orthosis, the soft lift assister for the knee.

Veale A J, Staman K, Van Der Kooij H. Soft, wearable, and pleated pneumatic interference actuator provides knee extension torque for sit-to-stand[J]. Soft Robotics, 2021, 8(1): 28-43.

Design of lower limb rehab robotics

● Actuators— Cable-driven (often used in soft exosuits)



[Actuation system]

(Harvard soft exosuits)]

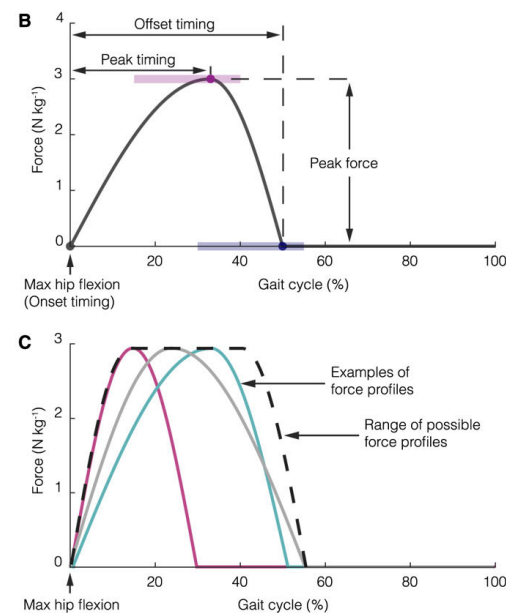
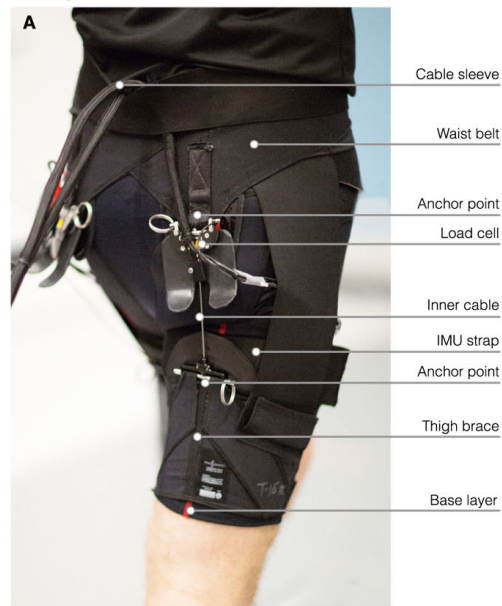


- **Cable-driven:** can apply controlled forces to the wearer by attaching at anchoring points in the wearable garment.
- Soft exosuits minimize the distal mass that is attached to the wearer through more proximally mounted actuation systems and flexible transmissions that transmit power to the joints.

Ding Y, Kim M, Kuindersma S, et al. Human-in-the-loop optimization of hip assistance with a soft exosuit during walking[J]. Science Robotics, 2018, 3(15): eaar5438.

Design of lower limb rehab robotics

● Case study— (soft exosuit)



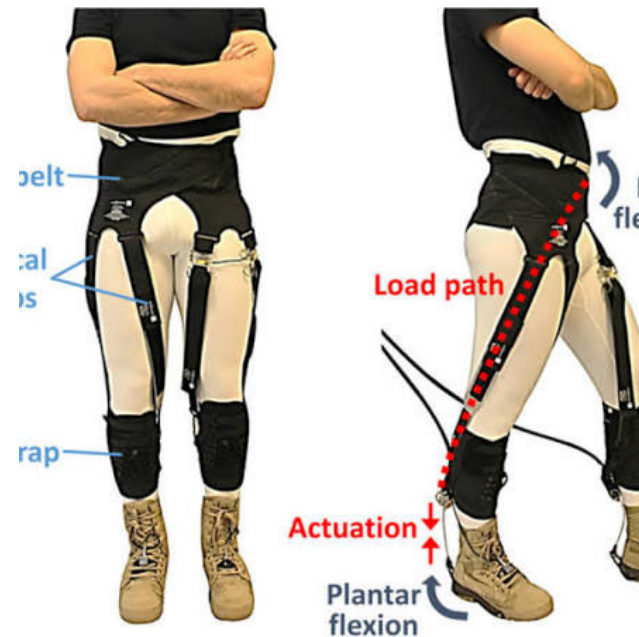
(Harvard soft exosuits)]

- **Hip soft exosuit:** A hip extension moment was generated by pulling the inner cable to create a tension between two anchor points.
- Parameterization of hip force profile. The hip force was chosen to be a combination of two parameterized sinusoidal curves joined at the peak.

Ding Y, Kim M, Kuindersma S, et al. Human-in-the-loop optimization of hip assistance with a soft exosuit during walking[J]. Science Robotics, 2018, 3(15): eaar5438.

Harvard soft exosuit (case study)

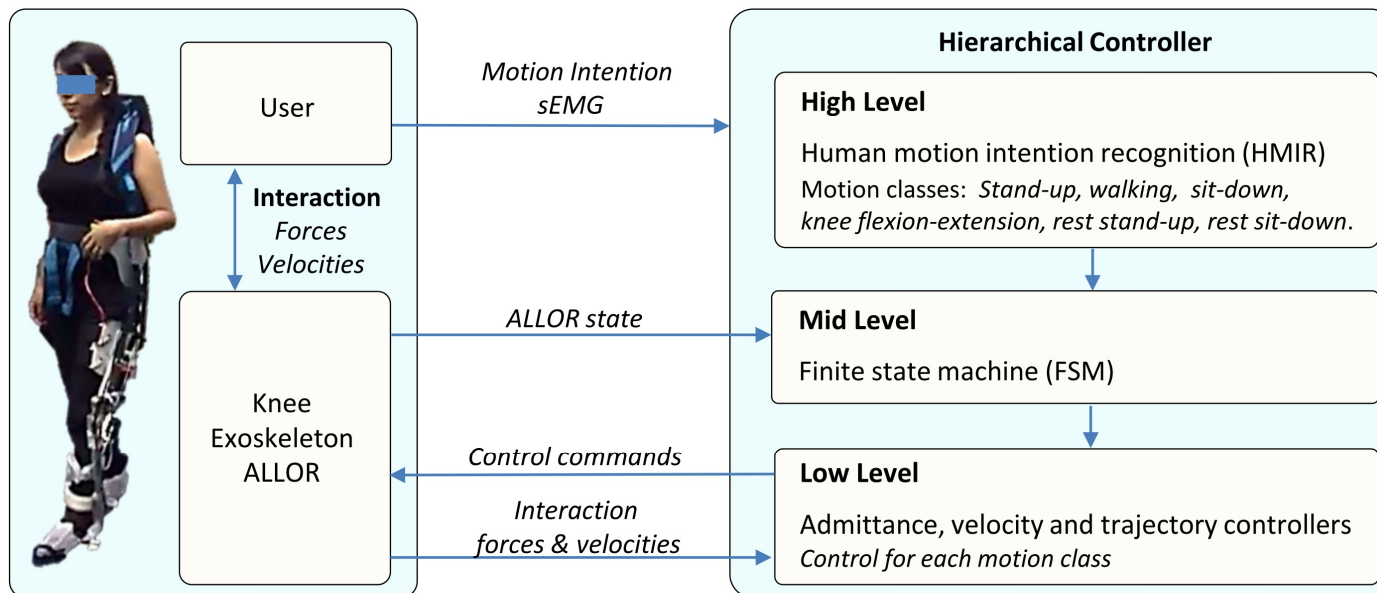
- Human locomotion assistances



- Harvard soft exosuit for mobility enhancing in real-world walking.

Control of lower limb rehab robotics

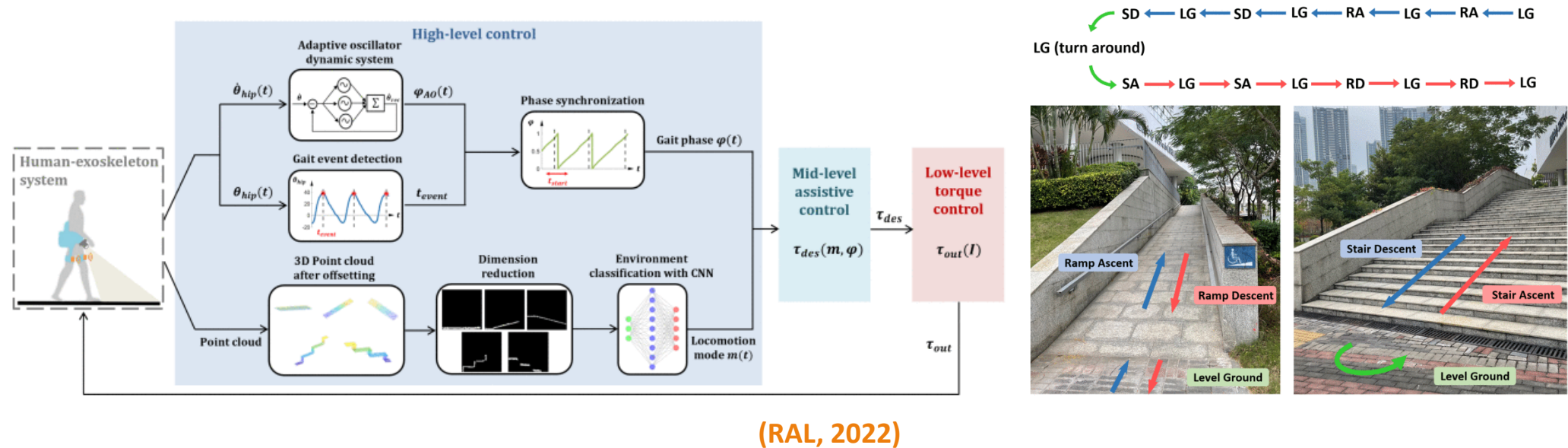
● Hierarchical Controller



- **Hierarchical controller:** consists of high-level (for human motion intention recognition), middle-level (for gait phase segmentation), and low-level control (for control commands execution)

Control of lower limb rehab robotics

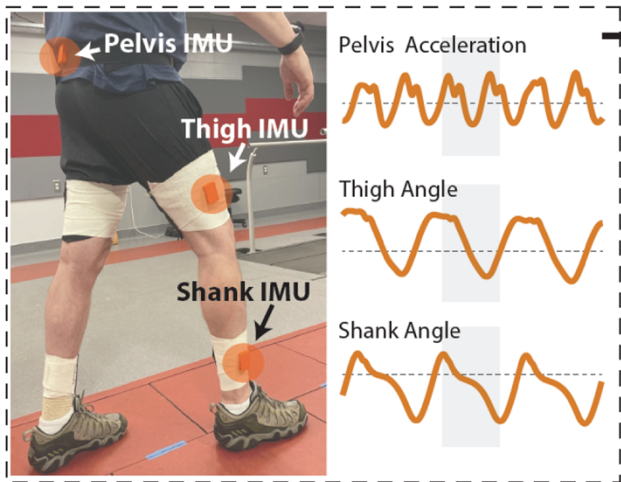
● High-level control (human motion intention recognition (HMIR))



- High-level control: mainly consisting of sensor for human motion intention detection, such as IMUs, depth-sensors, EMG, EEG

Control of lower limb rehab robotics

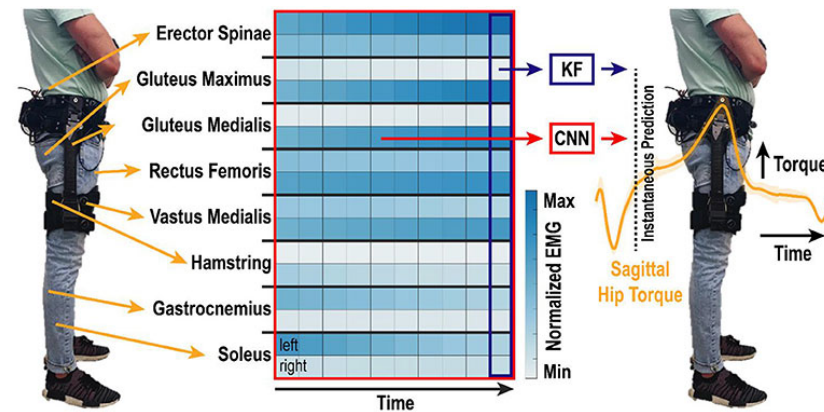
● High-level control (sensors used for human motion recognition)



(IMU, [Sensors, 2021])



(Depth-sensors, [RAL, 2022])

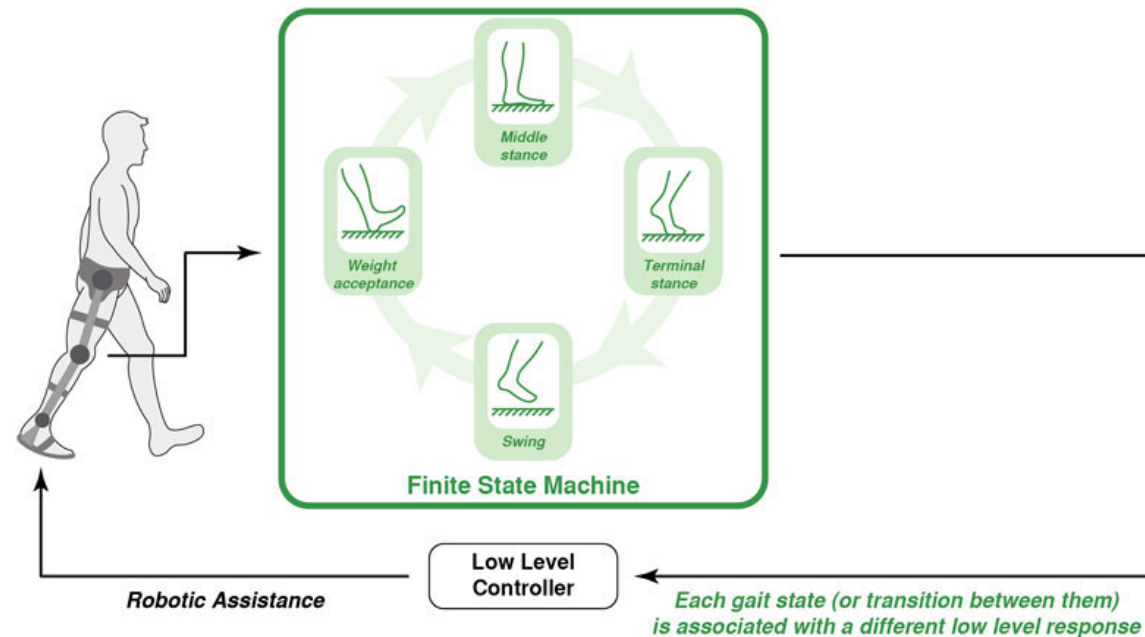


(EMG, [Fron. Robot, 2021])

- **High-level control:** mainly consisting of sensor for human motion intention detection, such as **IMUs, depth-sensors, EMG, EEG**

Control of lower limb rehab robotics

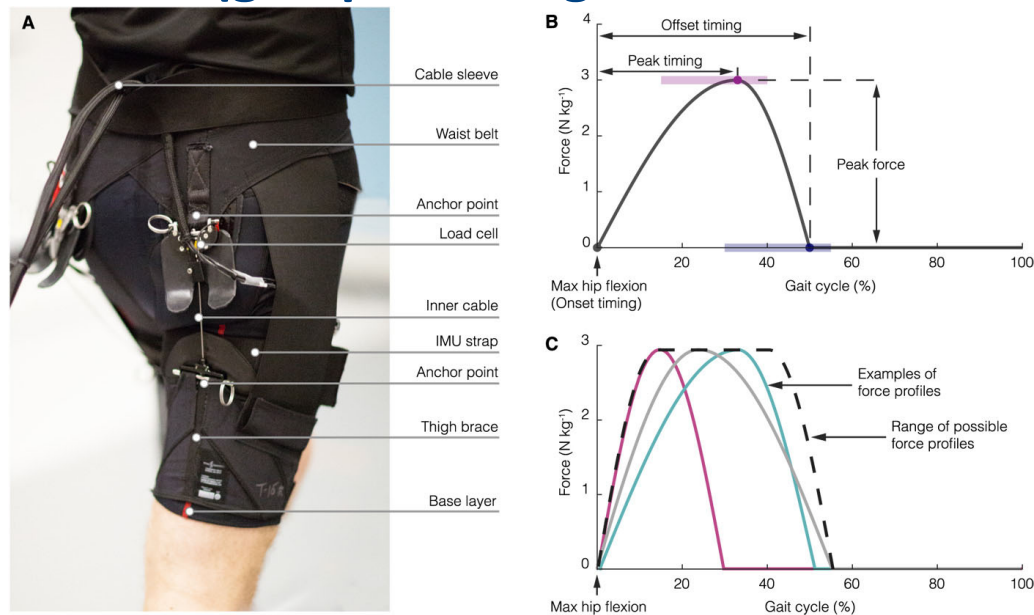
- Mid-level control (gait phase segmentation and assistive control)



- **Finite state machine:** each gait state is associated with a different low-level input torque commands and low-level response.

Control of lower limb rehab robotics

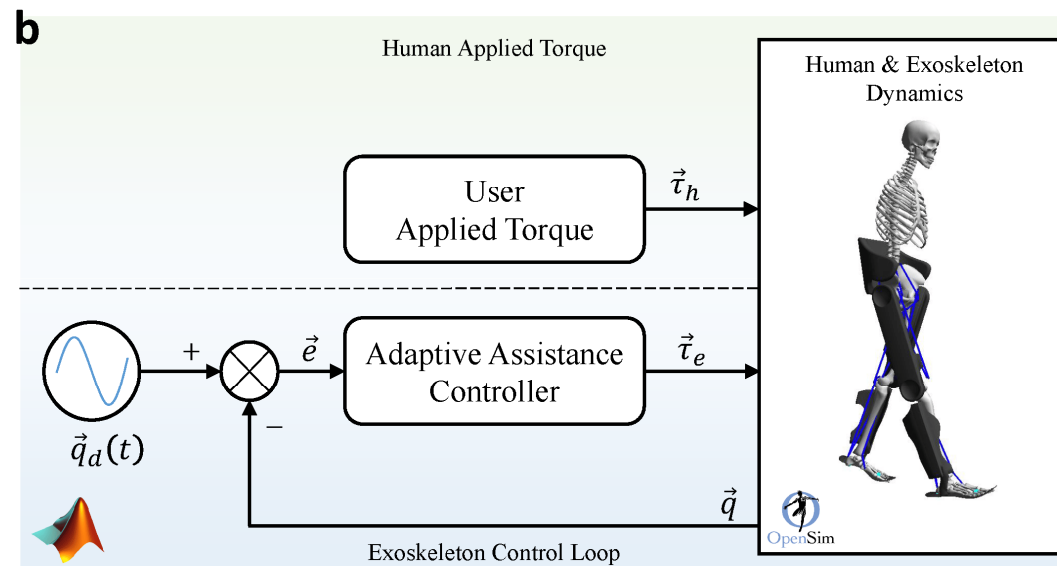
- Mid-level control (gait phase segmentation and assistive control)



- **Assistive control:** some mid-level controllers are designed for assistive torque generation for low-level controllers.

Control of lower limb rehab robotics

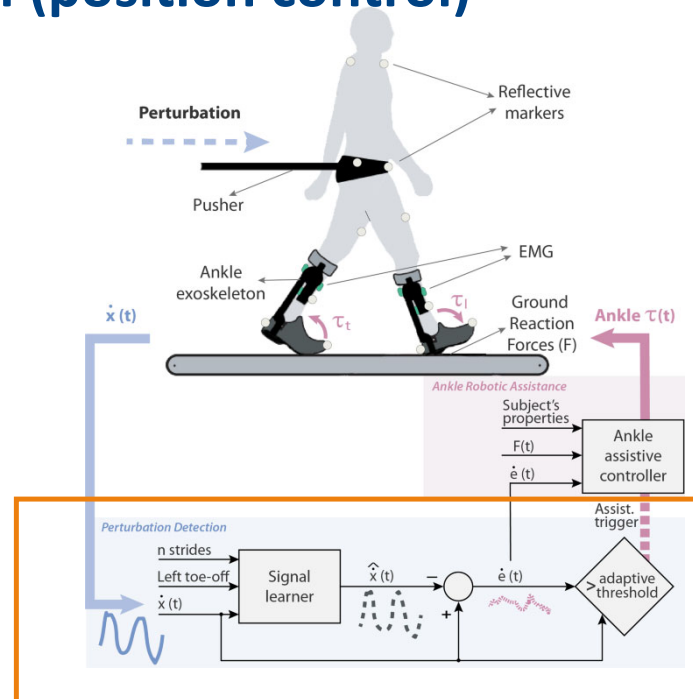
- Low-level control (torque control)



- **Torque control:** refers to the use of control algorithms and mechanisms that regulate the torque applied at the joints of the exoskeleton.

Control of lower limb rehab robotics

- Low-level control (position control)

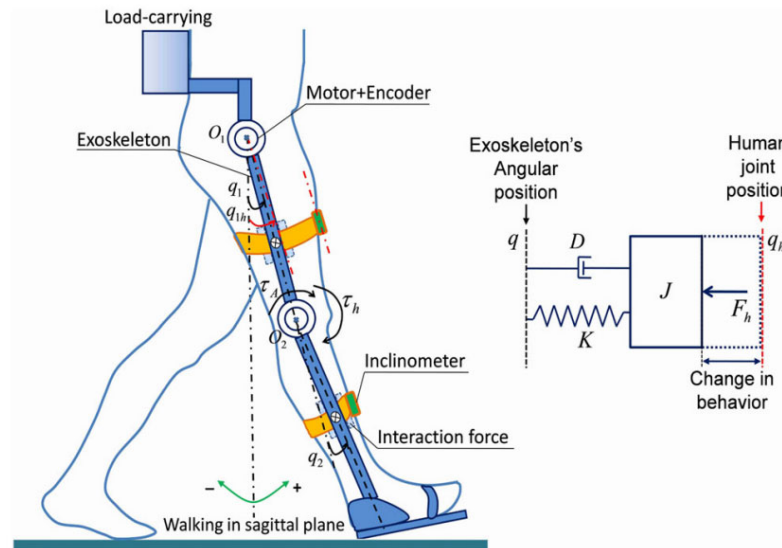


- **Position control:** a control strategy where the goal is to maintain specific joint angles or positions during movement.

Bayón C, Keemink A Q L, van Mierlo M, et al. Cooperative ankle-exoskeleton control can reduce effort to recover balance after unexpected disturbances during walking[J]. Journal of neuroengineering and rehabilitation, 2022, 19(1): 1-16.

Control of lower limb rehab robotics

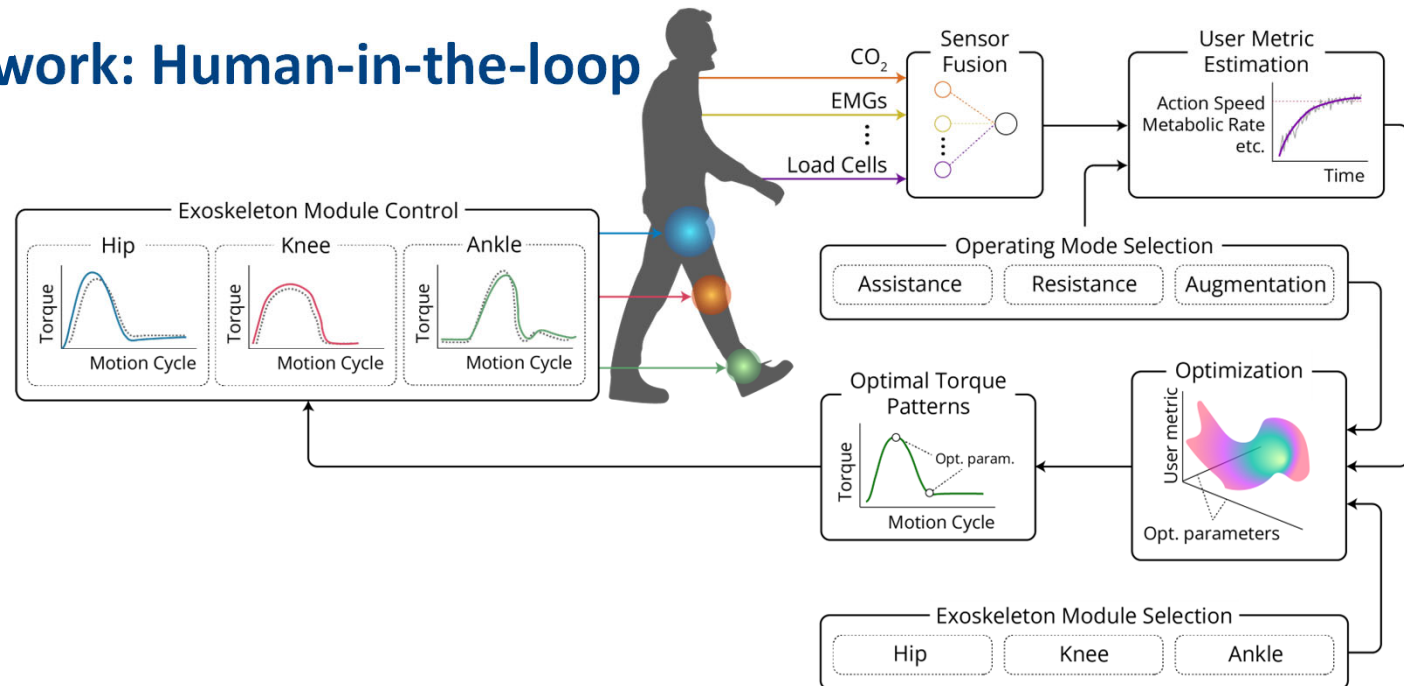
- Low-level control (compliance control)



- **Compliance control:** including impedance control and admittance control. This control strategy dynamically regulate force and position instead of controlling only force or position, which can provide comfortable, safe interaction between the users and exoskeletons.

Control of lower limb rehab robotics

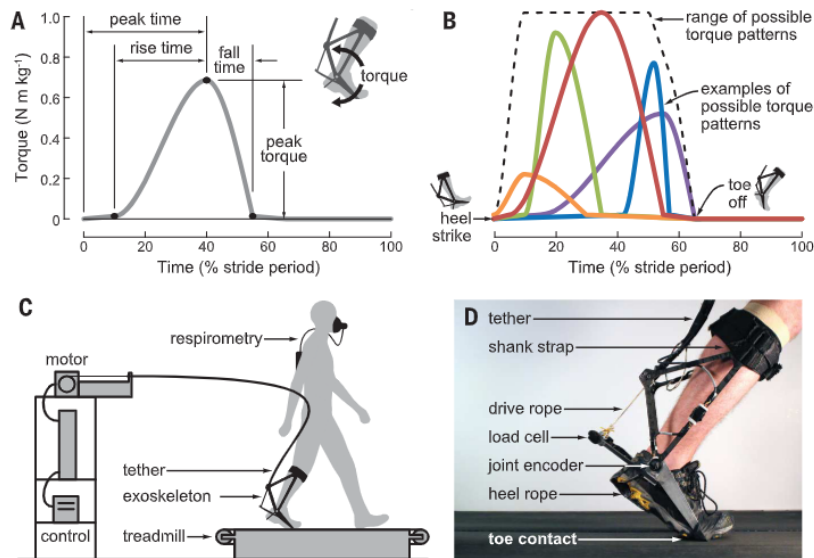
● Control framework: Human-in-the-loop



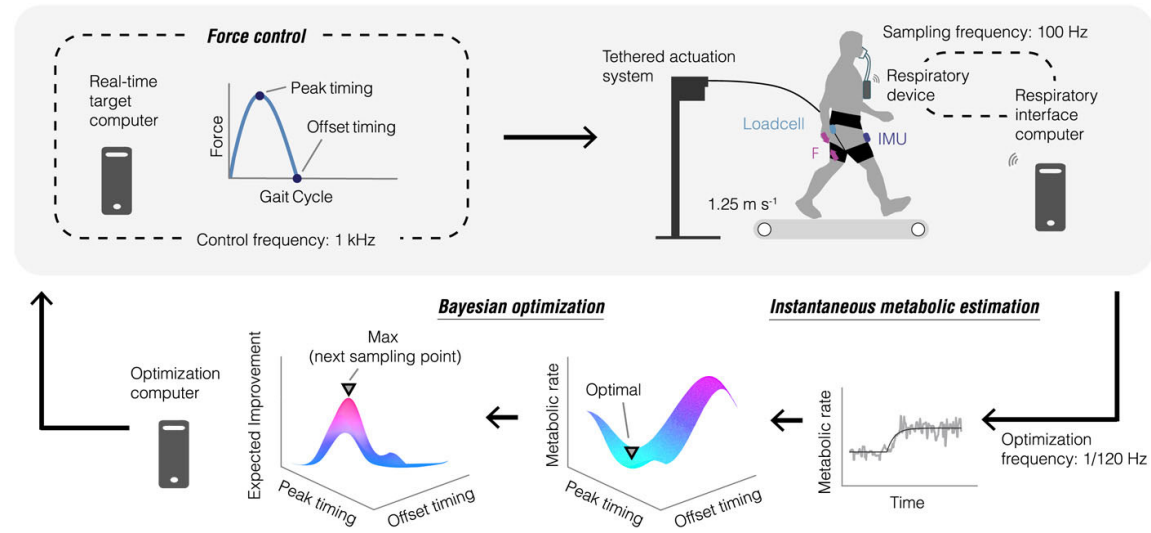
- **Human-in-the-loop:** refers to a control strategy where the exoskeleton's actions are influenced or guided by the user's intent and physiological signals in real-time. This approach involves active participation and collaboration between the exoskeleton's control system and the user, creating a closed-loop system that continuously adapts to the user's movements, intentions, and needs.

Control of lower limb rehab robotics

● Human-in-the-loop (Case study)



[Science, 2017]



[Science Robotics, 2018]

- **Science:** Optimized torque patterns from an ankle exoskeleton reduced metabolic energy consumption by $24.2 \pm 7.4\%$ compared to no torque.
- **Science Robotics:** Optimal peak and offset timing were found over an average of 21.4 ± 1.0 min and reduced metabolic cost by $17.4 \pm 3.2\%$ compared with walking without the device.

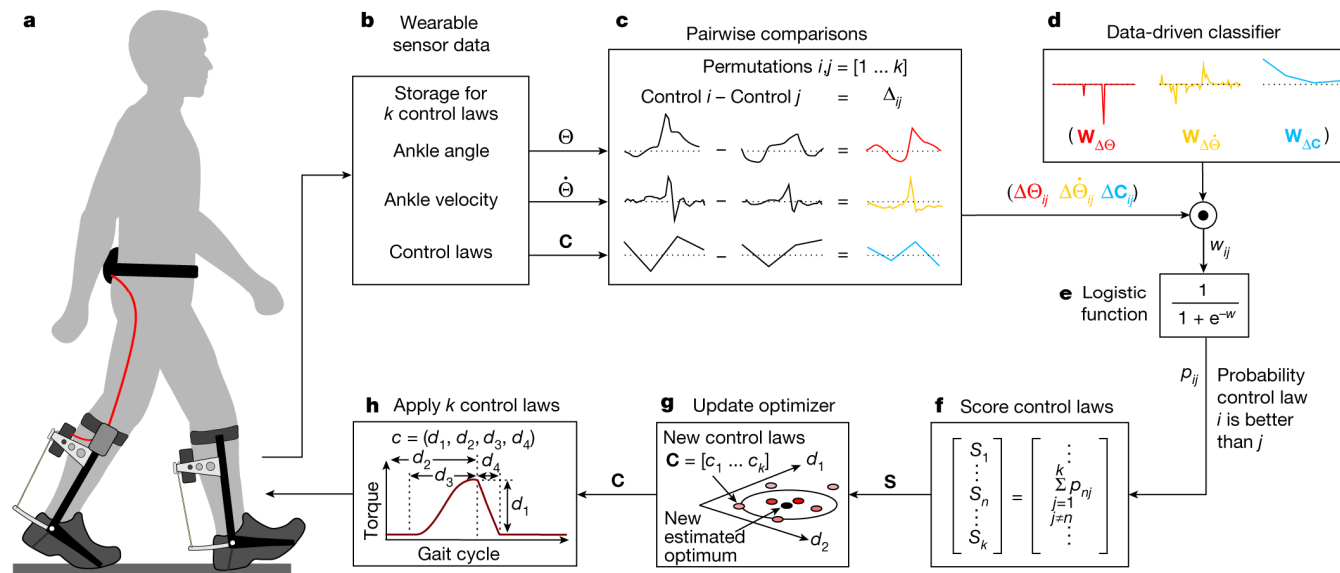
Zhang J, Fiers P, Witte K A, et al. Human-in-the-loop optimization of exoskeleton assistance during walking[J]. Science, 2017, 356(6344): 1280-1284.

Ding Y, Kim M, Kuindersma S, et al. Human-in-the-loop optimization of hip assistance with a soft exosuit during walking[J]. Science robotics, 2018, 3(15): eaar5438.

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Control of lower limb rehab robotics

● Control framework: Personalizing assistance



[Nature, 2022]

- **Personalizing assistance:** Personalized exoskeleton assistance provides users with the largest improvements in walking speed and energy economy. Using data-driven methods and optimization can achieve personalizing assistance in real-world walking.

Slade P, Kochenderfer M J, Delp S L, et al. Personalizing exoskeleton assistance while walking in the real world[J]. Nature, 2022, 610(7931): 277-282.

Stanford ankle exoskeleton-case study

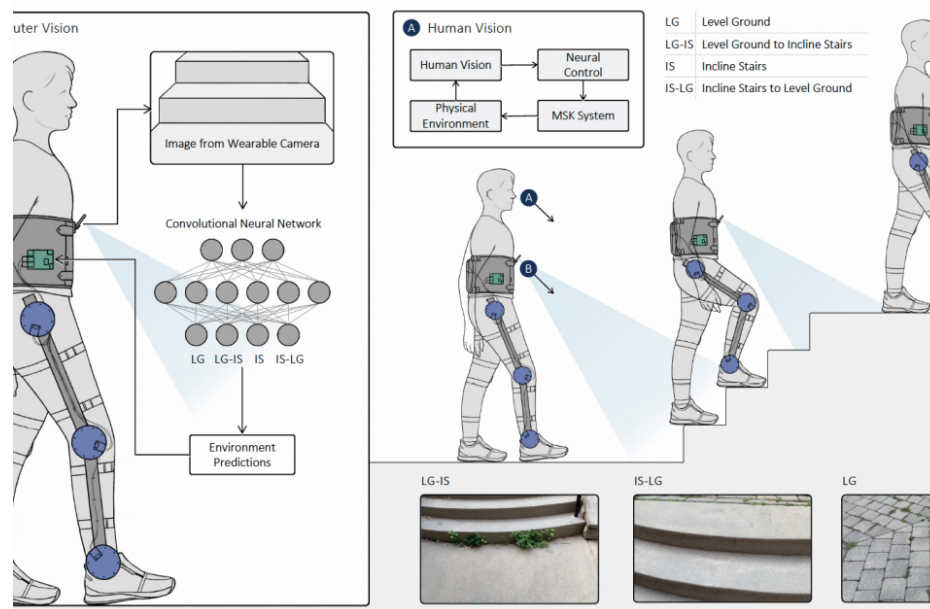
- Human locomotion assistances



- **Stanford exoskeleton** provides locomotion assistances to improve walking performance in real-world walking.

Advanced Topics in lower limb rehab robotics

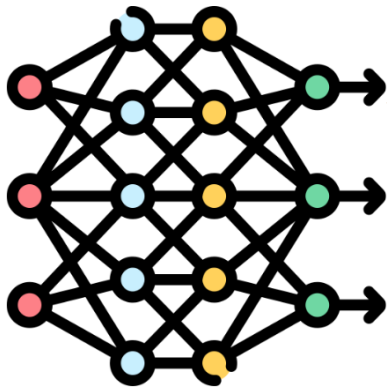
● Machine Vision



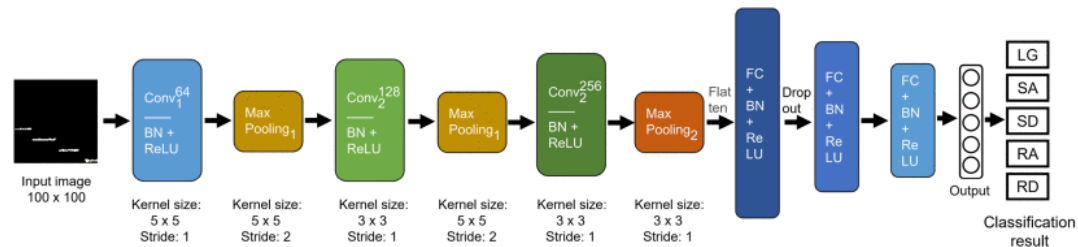
- **Machine vision** can be used to percept the environment and provide the environment information to the exoskeleton controllers, such as obstacle size, stair height.

Advanced Topics in in lower limb rehab robotics

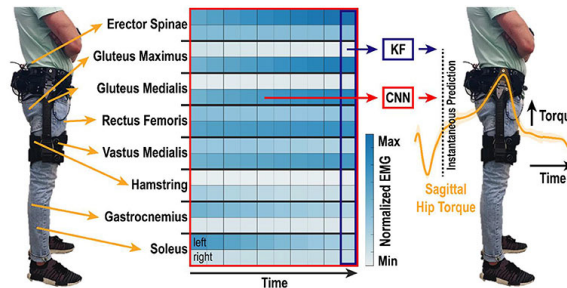
● Neural Network



Neural Network



(Neural network in depth-sensor data processing , [RAL, 2022])

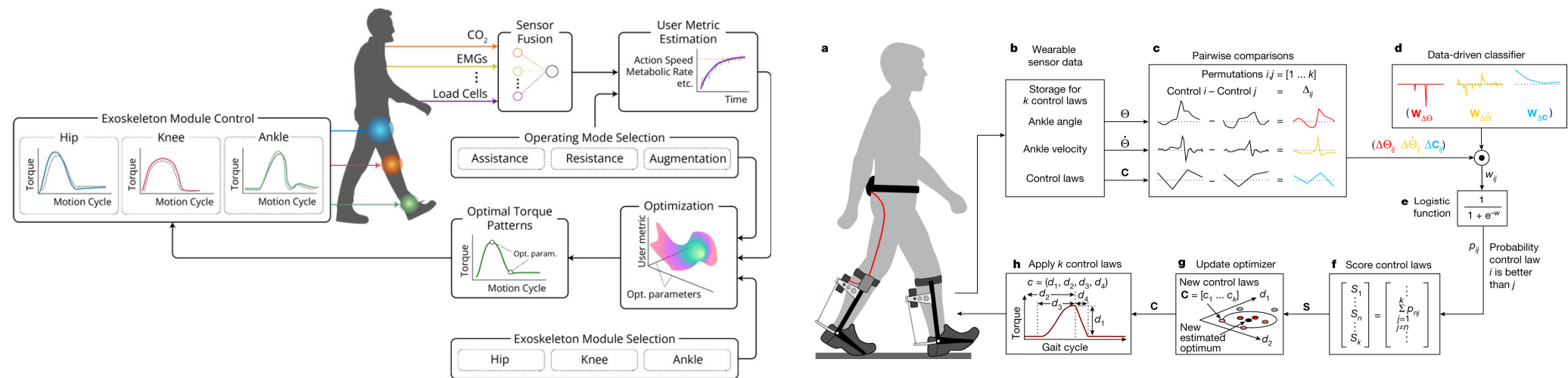


(Neural network in depth-sensor data processing , [Front. Robot., 2022])

- **Neural Network:** many technologies in exoskeletons involve neural network, such as the vision-system using CNN or RNN; EMG, EGG signal processing using time-delay neural networks (TDDNs) or attention-based neural networks.

Advanced Topics in in lower limb rehab robotics

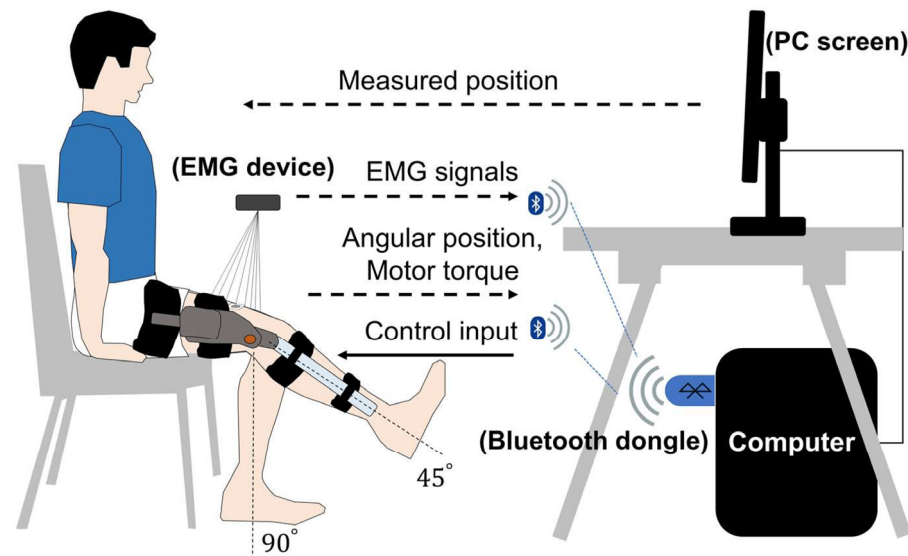
● Data-driven and Optimization



- **Data-driven and optimization:** Human in the loop control and personalize assistance control involve data-driven methods and optimization strategies for parameters tuning and energy consumption evaluation.

Advanced Topics in in lower limb rehab robotics

● Myoelectrical-interface



- **Myoelectrical-interface:** uses the electrical signals generated by muscles (myoelectric signals) to control the movements of the exoskeleton is capable of interpreting more complex muscle patterns to determine the user's intent.

Advanced Topics in in lower limb rehab robotics

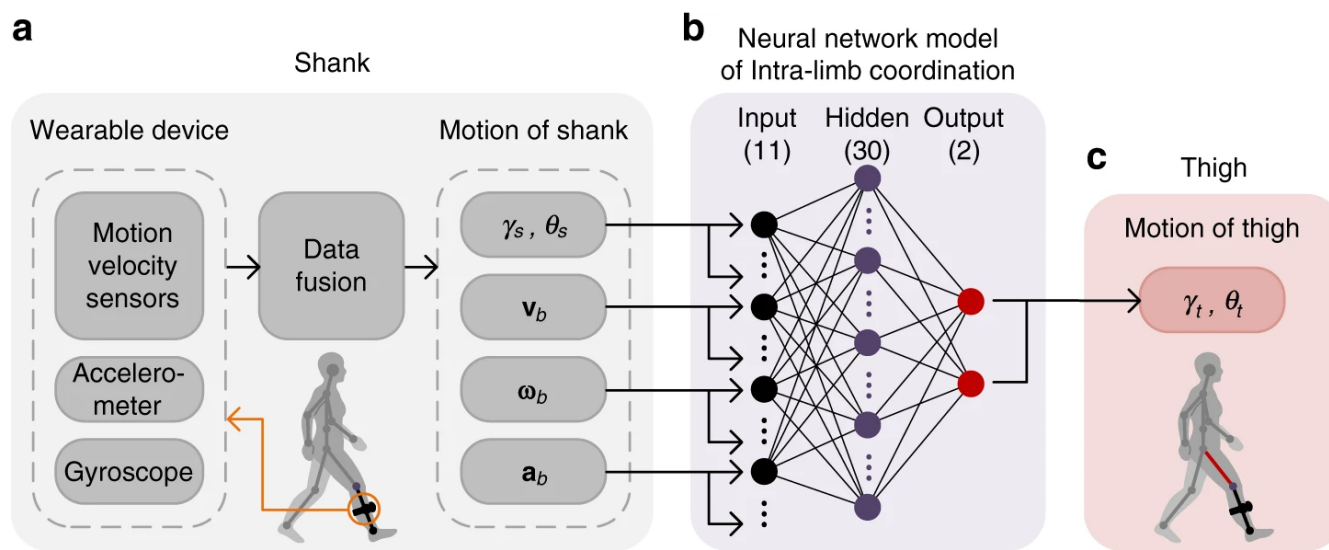
● VR/AR



- **VR/AR:** are increasingly being integrated into lower limb rehabilitation robotics to enhance the effectiveness and engagement of rehabilitation programs. These technologies provide immersive and interactive experiences, contributing to improved patient outcomes and motivation.

Advanced Topics in in lower limb rehab robotics

● Sensor Fusion



- **Sensor fusion:** AI can integrate data from multiple types of sensors (like IMU, force sensors, EMG, EEG, and cameras) to create a more holistic understanding of the user's movement and the state of the exoskeletons.

Lab experiments-case study

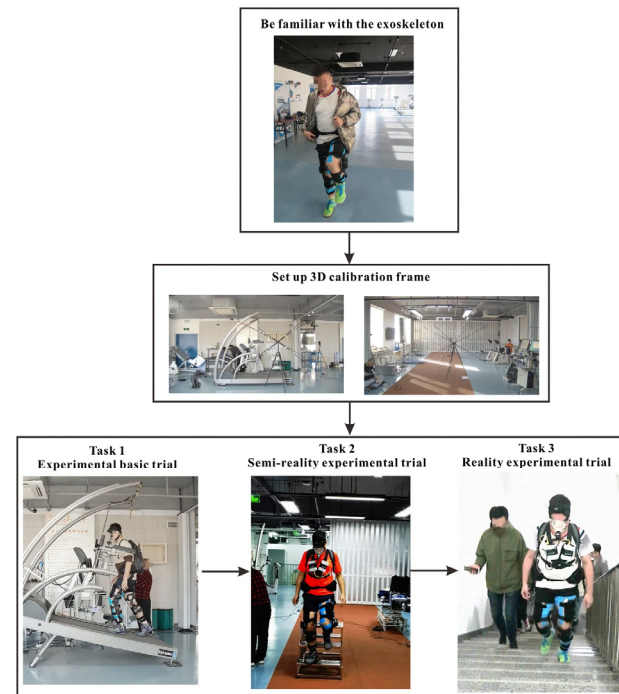
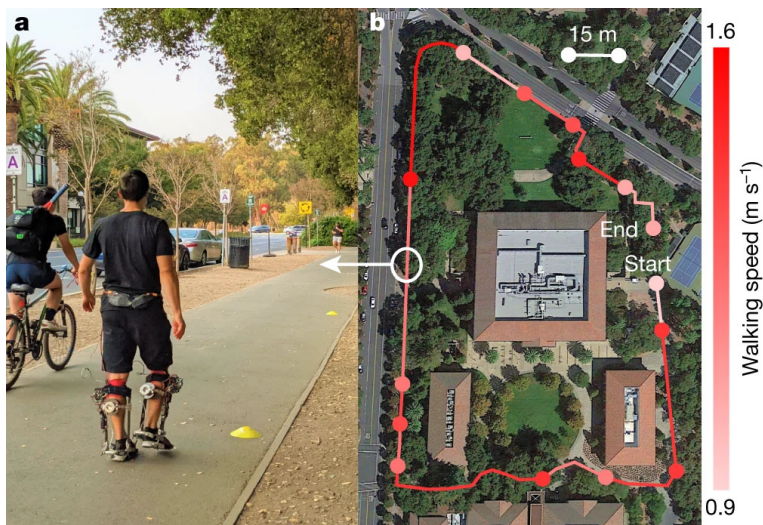
- Human locomotion assistances



- Building wearable robots to help people move better in various locomotion modes, such as varying walking speed, stair ascent/descent.

Advanced Topics in in lower limb rehab robotics

● Adaptive Control Strategies

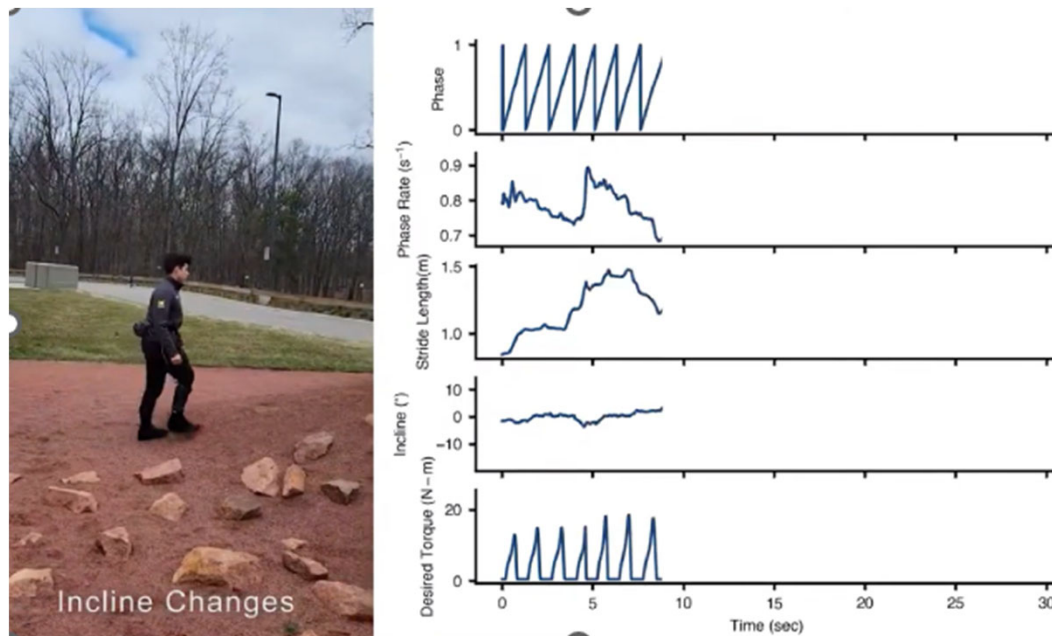


- **Adaptive control system:** AI adjust the exoskeleton's movements in real-time, adapting to changes in walking speed, terrain, and the user's gait.

Slade P, Kochenderfer M J, Delp S L, et al. Personalizing exoskeleton assistance while walking in the real world[J]. Nature, 2022, 610(7931): 277-282.

Example of adaptive control experiments

- Human locomotion assistances



- **Umich ankle exoskeleton** provides locomotion assistances to improve walking performance in real-world uneven terrains.

Clinical experiments

- Gait rehabilitation (stroke gait)



Locomat

- **Locomat:** The world's leading robotic medical device that provides highly repetitive physiological gait training - especially to severely impaired patients.

Clinical experiments

- Gait rehabilitation (stroke gait)

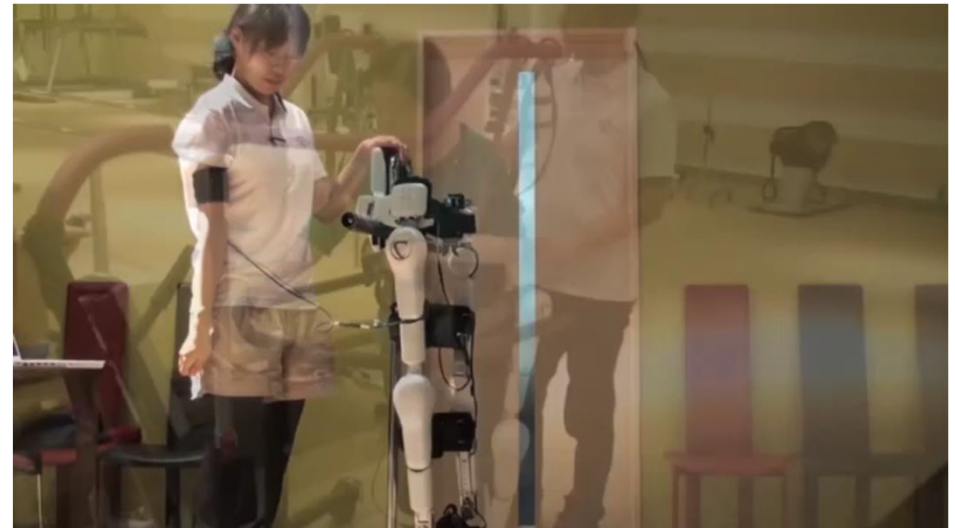


The NUS Portable Knee Ankle Robot

- Over-ground training; Patient starts, stops, and walks freely; Provide assistance for both knee and ankle joint

Clinical experiments

- Gait rehabilitation (stroke gait)



HAL exoskeleton (University of Tsukuba)

- **HAL exoskeleton:** internal sensors detect later weight shift; full-assistance for swing or assist-as-needed for swing.

Kawamoto H, Kamibayashi K, Nakata Y, et al. Pilot study of locomotion improvement using hybrid assistive limb in chronic stroke patients[J]. BMC neurology, 2013, 13(1):241.

Clinical experiments

- Gait rehabilitation (stroke gait)

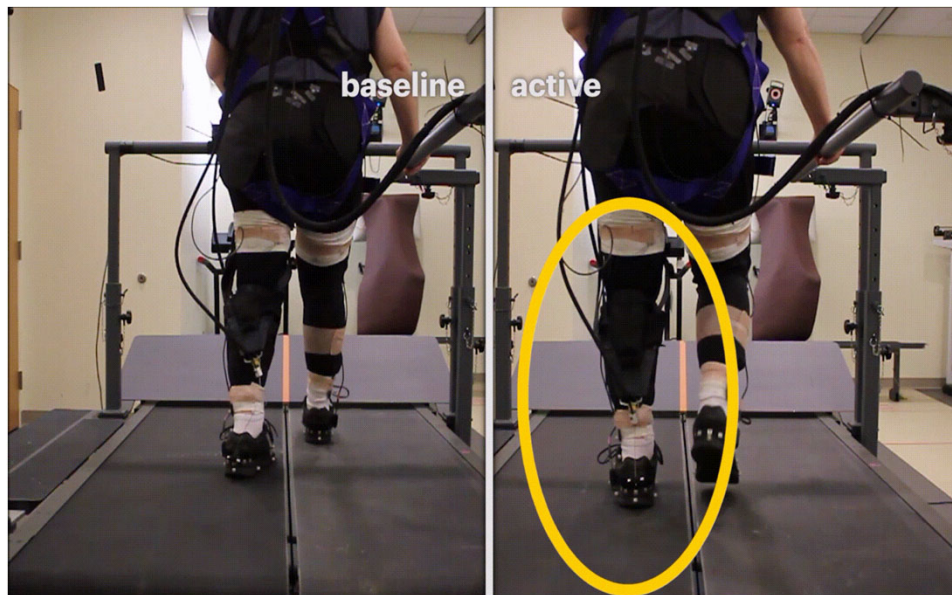


NUS Robotic Walker

- **Robotic walker:** active body weight support, balance training, integrated FES for joint muscle stimulation, best motor skill retention and functional outcome.

Clinical experiments

- Gait rehabilitation



Harvard soft exosuit

- **Soft exosuit:** Unilaterally supplement the paretic limb to generate both forward propulsion and ground clearance during walking.

Awad L N, Bae J, O'donnell K, et al. A soft robotic exosuit improves walking in patients after stroke[J]. Science translational medicine, 2017, 9(400): eaai9084.

Clinical experiments

- Gait rehabilitation



Toyota knee robot

- **Toyota knee robot for stroke:** simple construction and functions, including easy fitting and central touch panel operation; enhanced data management.

Awad L N, Bae J, O'donnell K, et al. A soft robotic exosuit improves walking in patients after stroke[J]. Science translational medicine, 2017, 9(400): eaai9084.

Challenges and future trends

● Challenge 1: Design (how to reduce cost)



\$300,000

Type	↕ Cost	↕ Notes	↕
Indego	\$80,000		
Ekso	\$100,000+	clinics only	
Hal 5	\$20,000	qualifying patients only	
Honda Walking Assist	\$375 monthly	bulk sale only - supports upper legs only	
SuitX Phoenix	\$40,000	test pilots only	
ReWalk	\$77,000		

- Design: high cost;
- High price of the motor, transmission, and sensors limit the commercialization of the powered exoskeletons

<https://costcharts.com/exoskeleton/>

Challenges and future trends

● Challenge 2: Clinical trials

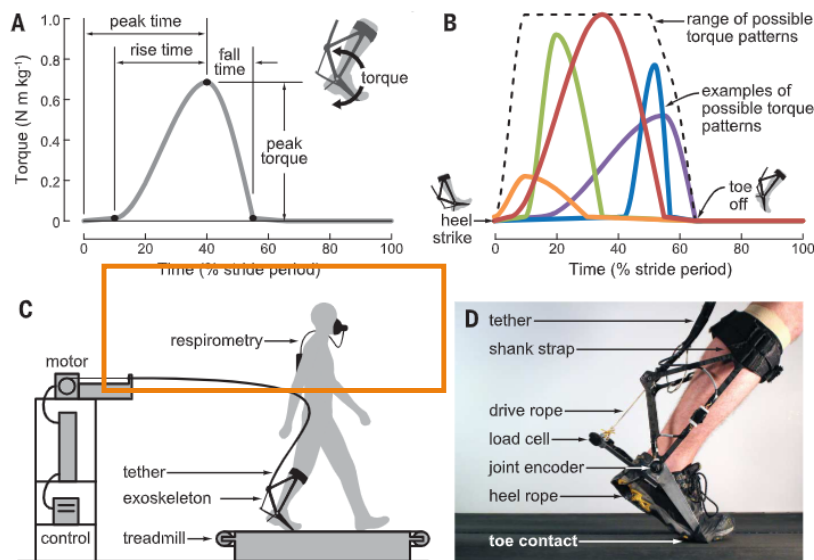


- Gait rehabilitation exoskeletons require a large number of subjects and clinical trials to evaluate the performance. This is a long and difficult process.

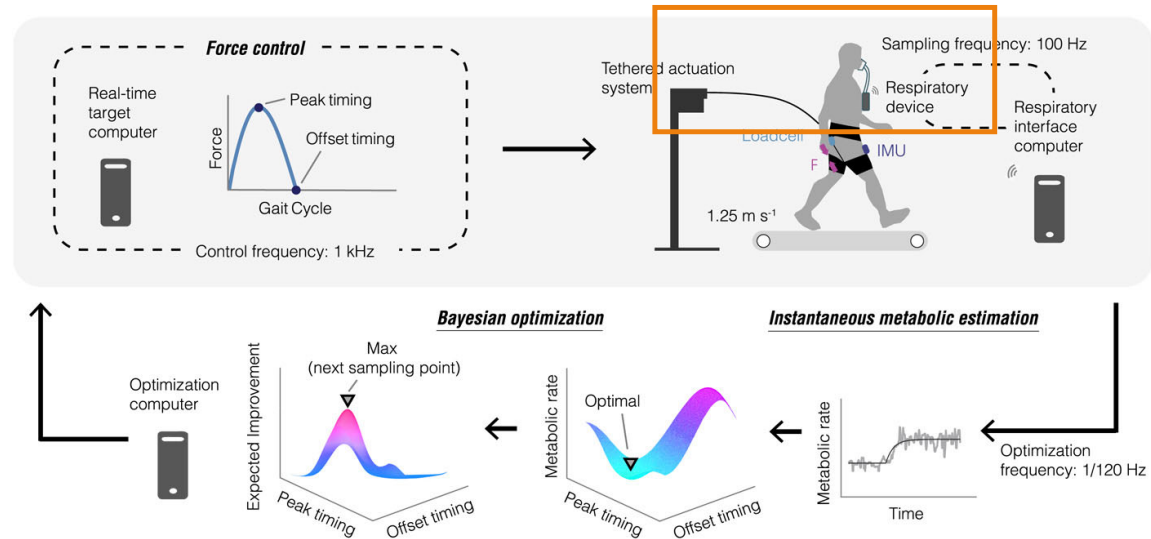
Apart from <https://eurecat.org/en/portfolio-items/able-human-motion/>

Challenges and future trends

● Challenge 3: Accurate metabolic-cost estimates



[Science, 2017]



[Science Robotics, 2018]

- Accurate metabolic-cost estimates have to be implemented as real-time input to exoskeleton-control schemes to better prescribe assistance and to maximize individual metabolic-cost reductions.

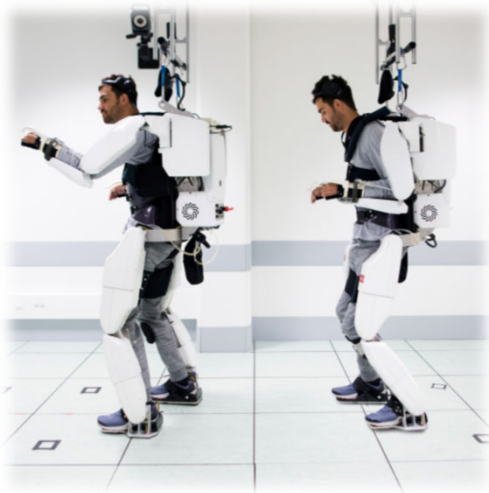
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Ding Y, Kim M, Kuindersma S, et al. Human-in-the-loop optimization of hip assistance with a soft exosuit during walking[J]. Science robotics, 2018, 3(15): eaar5438.

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Challenges and future trends

● Challenge 4: Ensuring user safety



(risk of fall without of the safety hook)



(risk of spraining back)

- Ensuring user safety is a key exoskeleton-design consideration for unsupervised or remotely supervised community-based or home-based gait training. Unlike upper-extremity training, gait training inherently incurs a fall risk. Importantly, for gait training with a powered exoskeleton, wearer instability must be limited in the case of device failure.

Challenges and future trends

● Future trends 1: Collaborations between clinicians and exoskeleton designers



- **Collaborations:** will enable targeted case studies with a small number of selected participants to inform biological mechanisms that warrant larger clinical trials and drive changes in functional outcomes, to enable the design of optimization algorithms for individualized assistance.

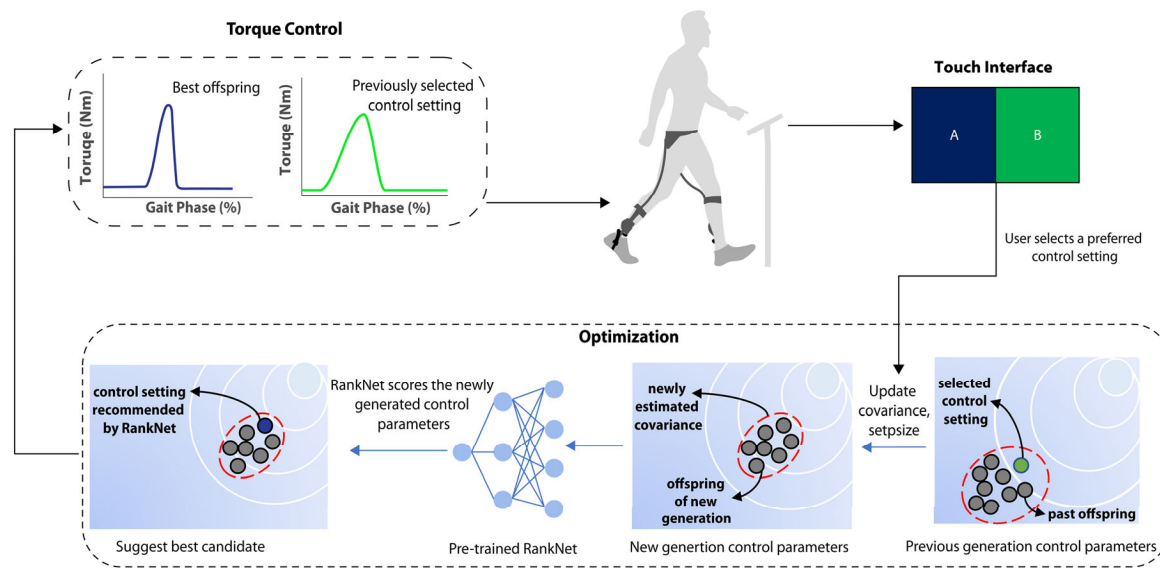
Apart from <https://www.ablehumanmotion.com/able-exoskeleton-provides-a-more-physiological-gait-pattern/>



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Challenges and future trends

● Future trends 2: User preference optimization



[Science Robotics, 2023]

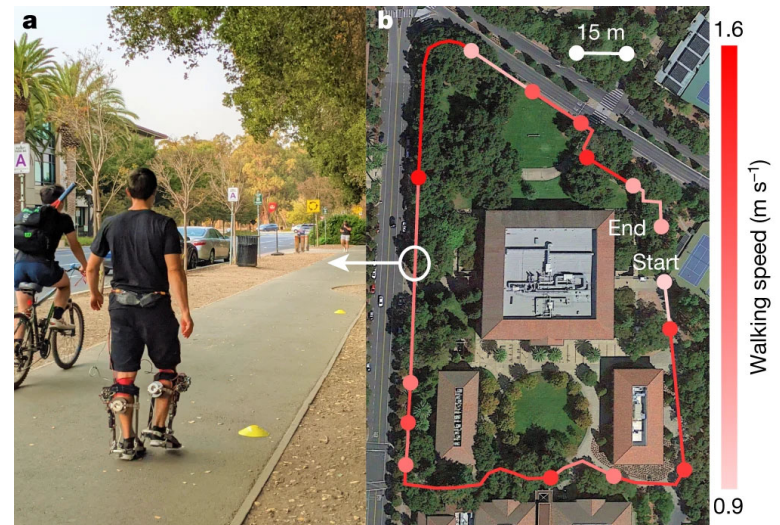
- **User preference optimization:** Once people wearing the powered prosthesis, they are trying to comfortably control their exoskeletons. This user preference is not considered in current studies. Future prosthesis controllers may involve designing controller to **maximize user preference**.

Challenges and future trends

● Future trends 3: Assistances in real-world walking



(Harvard soft exosuit)



(Stanford exoskeleton) [Nature, 2022]

- **Assistance in real-world walking:** While some exoskeletons enable outdoor walking assistance, the vast majority of research is currently done in the lab. Exoskeletons will eventually be used in outdoor environments, and stroke patients wearing exoskeletons to assist them in walking outdoors.

Challenges and future trends

● Future trends 4: Connection between Large Language Models and exoskeletons



- **LLM in exoskeletons:** AI makes large language models an important bridge between humans and wearable robotics. They can translate human commands into a language that machines understand and convert machine feedback into a language comprehensible to humans. This interactive approach greatly enhances the usability of robot, allowing them to better integrate into our work and life.

Siviy C, Baker L M, Quinlivan B T, et al. Opportunities and challenges in the development of exoskeletons for locomotor assistance[J]. Nature Biomedical Engineering, 2023, 7(4): 456-472.



THANK YOU